Study of the resolution of mass and heat transfer for large-scale simulation of fire propagation on solid fuel plates

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1 Presentation of the Organization

1.1 The ASNR

The French Nuclear Safety and Radiation Protection Authority (**ASNR**) is an independent French administrative authority established on January 1, 2025. Operating on behalf of the State, it regulates civil nuclear activities in France. It was formed through the merger of two entities: the Nuclear Safety Authority (ASN) and the Institute for Radiological Protection and Nuclear Safety (IRSN). Both ASN and IRSN originally emerged from the lessons learned following the 1986 Chernobyl accident.

The main missions of ASNR are to:

- **Conduct research**, both internally and through partnerships with other French or international research organizations.
- **Assess the safety** of civil nuclear installations at every stage of their lifecycle, from design to decommissioning.
- **Regulate** by developing and enforcing technical regulations, evaluating authorization requests for facilities and transport, and issuing nuclear permits.
- **Inspect**: ensuring that every party responsible for a nuclear activity fulfills its obligations regarding nuclear safety and radiation protection. Control actions take the form of formal decisions, requirements, field inspections, and, where necessary, sanctions.
- **Manage radiological emergency situations**, assessing the nature and severity of the event, its progression, potential developments, and both proven or potential radiological consequences.
- **Provide training**: offering a wide range of training courses in nuclear safety and radiation protection.
- **Inform and engage in dialogue**: ASNR informs the general public about the status of nuclear safety and radiation protection in France.
- **Develop a radiation protection culture** among citizens.

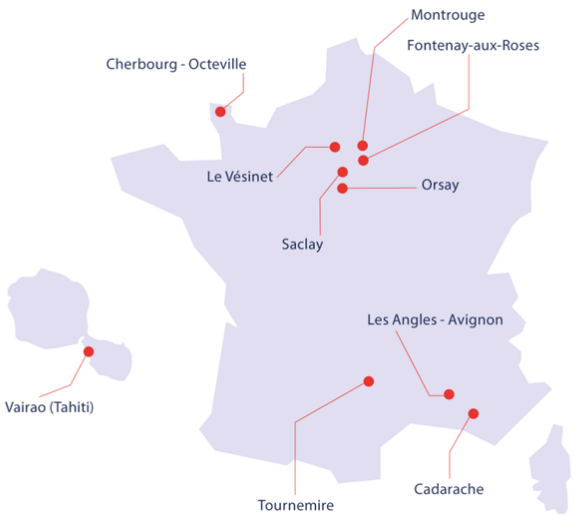


Figure 1. Location of ASNR sites in France

ASNR operates about ten sites across France, as shown in Figure 1, allowing it to carry out its regulatory and monitoring missions in close proximity to nuclear facilities. Its headquarters are located in Montrouge (Hauts-de-Seine), south of Paris. In addition to this headquarters, ASNR maintains regional divisions and technical facilities near major nuclear sites (such as Chinon, Civaux, Bugey, Tricastin, La Hague, etc.).

1.2 The CEA Cadarache Center

ASNR is established within the CEA Cadarache center, created in 1959. Located approximately forty kilometers north of Aix-en-Provence, this center spans 1,600 hectares and accommodates roughly 5,500 personnel daily, including 350 ASNR staff members. ASNR's presence at Cadarache includes a site management department, attached to the Heritage and Territory division, as well as several units belonging to three technical divisions and functional departments. It houses a significant portion of technical and scientific teams notably from the former IRSN as well as several specialized expertise laboratories focused on reactor safety, accident analysis, and material behavior in irradiated environments.

1.3 Fire and Explosion Laboratory (LIE)

Within ASNR, the Internal Hazards and Industrial Risks Department (SA2I) encompasses three laboratories dealing with fire and explosion issues: the Fire and Explosion Expertise Bureau (BEXIE), which handles safety appraisals; the Fire Experimentation Laboratory (LEF), which conducts experimental work; and the Fire and Explosion Laboratory (LIE), located at Cadarache, which specializes in the numerical modeling of fire and explosion risks in nuclear installations. It is within this laboratory (LIE) that I completed my internship.

The laboratory consists of 13 permanent engineers and researchers, 3 PhD candidates, 2 interns, and 2 work-study apprentices, divided between two primary research themes:

- fire modeling;
- explosion modeling.

2 Introduction

Accurately capturing turbulent combustion, thermal radiation, and buoyant flow dynamics in wall-bounded reactive flows is vital for predicting flame spread over solid combustible surfaces [1]. Establishing the numerical resolution demands and grid fidelity guidelines for such configurations is essential prior to modeling large-scale flame propagation across plates and slabs. This work provides a systematic numerical investigation progressing from fundamental non-reactive buoyant cavities to complex reactive vertical wall fires using Large Eddy Simulation (LES). The study begins with a non-reactive Differentially Heated Vertical Cavity (DHVC) benchmark, constructing a comprehensive sensitivity matrix to evaluate grid convergence, spatial discretization schemes, and core mesh refinement across a spectrum of low-to-high Rayleigh numbers (Ra), thereby establishing baseline numerical efficiency, stability limits, and computational cost scaling. Building upon these non-reactive insights, the investigation transitions to a reactive vertical wall fire configuration, where wall-normal grid refinement studies are conducted to evaluate boundary layer dynamics in viscous units (y^+), mean velocity and thermal profiles, turbulent fluctuation variances, wall heat flux partitioning validated against experimental data, and LES quality indices. Ultimately, this work quantifies the balance between numerical accuracy and computational expense, delivering a validated framework for future simulations of flame propagation such as complementing the ongoing experimental work at the CISCCO facility[7] and for simulations over plate and slab geometries.

3 Literature Review

The numerical modeling of wall-bounded buoyant flows requires a systematic understanding of fluid transport spanning from canonical non-reacting natural convection to complex reacting turbulent boundary layers. Predicting momentum and heat transfer along vertical surfaces presents computational challenges due to steep velocity gradients, strong buoyant force coupling, variable fluid properties, and near-wall resolution constraints. This review synthesizes key foundational studies across canonical natural convection, turbulent vertical wall fires, convective heat transfer modeling, and the underlying scaling physics required for predictive modeling.

3.1 Canonical Buoyant Flows and Boundary Layer Physics

The theoretical foundation for buoyancy-driven boundary layer modeling relies on benchmark Direct Numerical Simulations (DNS) of non-reacting natural convection. Versteegh and Nieuwstadt [2] established the DNS dataset for natural convection between two infinite vertical differentially heated planes at Rayleigh numbers up to $\mathrm{Ra} = 5.0 \times 10^6$.

To fully interpret these benchmarks physically, the distinct hydrothermal structure that develops inside differentially heated vertical cavities (DHVC) must be understood. Unlike forced internal flows driven by external pressure gradients, natural convection is self-sustained by volumetric buoyancy forces: fluid adjacent to the hot wall experiences a density deficit and accelerates upward, while fluid near the cold wall accelerates downward. This creates thin, highly active boundary layers enclosing a relatively quiescent, stably stratified central core.

However, while DNS provides exact physical descriptions, its computational cost scales excessively with the Rayleigh number, making it unfeasible for engineering-scale configurations or reacting environments. This limitation necessitates Large Eddy Simulation (LES), where large energy-carrying eddies are explicitly resolved while small, universal scales are modeled. Within this framework, the energy cycle is governed by a direct competition between buoyancy production and turbulent dissipation:

- **Buoyancy-Driven Velocity Maxima:** Unlike forced convection where wall shear stress (τ_w) is governed by freestream momentum or external pressure gradients, natural convection generates an internal velocity peak (W_{max}) within the near-wall boundary layer due to localized density deficits.
- **Turbulent Kinetic Energy Budgets and Dissipation Sinks:** Exact evaluation of TKE budgets demonstrated that buoyancy acts simultaneously as a driver of vertical fluid acceleration and a generator of turbulent shear mixing near the wall. Kinetic energy produced by buoyancy is transferred across the spectrum and ultimately destroyed by viscous dissipation (ϵ), serving as the physical ”fingerprint” of small-scale turbulence intensity.
- **Numerical Solver Benchmarks:** The precise spatial profiles of mean velocity, temperature, and Reynolds stress tensors provided by Versteegh and Nieuwstadt [2] serve as the primary validation standard for CFD frameworks in Differentially Heated Vertical Cavity (DHVC) configurations prior to integrating reacting aspects.

3.2 Large Eddy Simulation of Turbulent Vertical Wall Fires

Extending boundary layer models from closed, non-reacting cavities to open-atmosphere vertical wall fires introduces complex reacting shear layers, fuel injection dynamics, and intense localized heat release. Ren et al. [1, 6] performed Large Eddy Simulations (LES) of turbulent vertical wall fires supplied with gaseous fuel through porous wall burners, while utilizing the underlying

experimental data from de Ris et al. [5] to compare and interpret the reactive boundary layer and heat transfer results.

From a physical perspective, transitioning to reacting boundary layers introduces severe modeling hurdles for Subgrid-Scale (SGS) formulations. Standard models (like Smagorinsky) originally tuned for isothermal wall flows often over-dissipate energy in buoyant plumes because they fail to account for thermal expansion, flame-induced local acceleration, and turbulent backscatter (the reverse transfer of energy from small to large scales). Advanced SGS treatments such as the Wall Adapting Local Eddy viscosity (WALE) model [4] correctly recover proper near-wall scaling behavior without requiring artificial damping functions.

Key physical insights from their work include:

- **Flame Attachment and Buoyant Acceleration:** Gaseous fuel pyrolyzed or injected at the wall reacts within a thin diffusion flame zone. The resulting extreme temperature rise ($\Delta T > 800$ K) drives violent vertical buoyancy forces, controlling core velocity acceleration and altering wall shear stress dynamics.
- **Coherent Structure Resolution:** The LES framework effectively captured transient, large-scale turbulent structures responsible for entraining surrounding air and dictating flame stability along the vertical boundary.

3.3 Near-Wall Convective Heat Transfer and Grid Resolution Constraints

The characterization of near-wall turbulent structures depends heavily on the local wall Reynolds number (Re_τ), which dictates the momentum and thermal energy transfer mechanisms within the inner boundary layer. In large-scale fire safety simulations, fully resolving the viscous sublayer ($y^+ < 1$) across extended wall surfaces while maintaining appropriate wall Reynolds number scaling is computationally prohibitive. Consequently, coarsened grids are often employed near boundaries, requiring accurate wall function formulations to predict local wall heat flux ($q_w = k \left.\frac{\partial T}{\partial y}\right|_{\text{wall}}$) and skin friction ($C_f$).

Ren and Wang [1] systematically evaluated convective heat transfer models for LES fire applications, identifying severe limitations in traditional log-law formulations:

- **Breakdown of Standard Wall Functions:** Classical logarithmic law-of-the-wall models rely on constant fluid properties and negligible volumetric source terms near the boundary. In fire plumes, intense thermal gradients invalidate constant-property assumptions.
- **Variable Transport Properties:** Near the hot wall surface, dynamic viscosity (μ) increases non-linearly with temperature via Sutherland's law, while density (ρ) decreases significantly. Ren and Wang [1] emphasized that wall shear stress (τ_w) and friction velocity ($u_\tau = \sqrt{\tau_w/\rho_w}$) must be evaluated strictly using fluid properties evaluated at the actual wall temperature (T_{wall}) to avoid artificial distortion of non-dimensional parameters.
- **Nusselt Number Reformulation:** Evaluating local Nusselt numbers (Nu_z) using ambient temperature differences ($\Delta T = T_{\text{wall}} - T_\infty$) produces unrealistically inflated values. The true thermal driving force across the near-wall layer is governed by the flame core temperature, requiring $Nu_z = \frac{z \cdot |\partial T/\partial y|_w}{T_{\text{max}} - T_{\text{wall}}}$.

3.4 Scaling Frameworks and Energy Partitioning in Buoyant Boundary Layers

To meaningfully evaluate LES predictive capability against reference DNS or experimental benchmarks across varying mesh topologies, raw physical values must be placed into a consistent non-dimensional scaling framework. In buoyancy-driven systems, direct comparisons

of dissipation peaks or turbulent kinetic energy can be skewed by mesh-dependent energy redistribution.

Understanding how resolved kinetic and thermal energy partitions relative to subgrid-scale (SGS) contributions is essential for assessing model fidelity. By implementing rigorous scaling prefactors , researchers can decouple numerical grid artifacts from true physical phenomena. This ensures that variations observed in grid sensitivity studies reflect genuine turbulence resolution dynamics rather than arbitrary dimensional scaling mismatches, providing a clear pathway for validating LES models against canonical benchmarks.

In summary, the literature underscores a clear methodological progression: while canonical differentially heated cavities rely on high-fidelity DNS benchmarks to understand basic buoyant boundary layers and energy dissipation, extending these principles to complex vertical wall configurations and turbulent fires introduces severe challenges in near-wall resolution and subgrid-scale modeling. Accurately predicting wall heat transfer and boundary layer development requires careful handling of variable property effects and SGS closures suited for buoyant plumes. Building upon these insights, the present work systematically evaluates the sensitivity of LES predictions spanning numerical schemes, SGS models, and grid topologies to establish a reliable framework for wall-bounded buoyant flows.

4 Numerical Model for Wall-Bounded Buoyant Flows

4.1 Governing Equations and Low-Mach LES Framework

The continuous governing equations modeling the fluid hydrodynamics, thermal transport, and chemical-thermal interactions within the computational domain are established using a Large Eddy Simulation (LES) framework. Following the mathematical formulation implemented in the CALIF^{3}S software environment, the conservation equations are Favre-filtered to isolate large resolved structures from subgrid-scale (SGS) features [3].

Because fire compartment cases involve extreme thermal loads and large density variations, the solver employs an asymptotic expansion in powers of the Mach number ($M = ||\mathbf{v}||/c \to 0$). This low-Mach formulation filters out acoustic wave propagation while fully accounting for real volumetric expansions ($\nabla \cdot \mathbf{v} \neq 0$). The pressure field is decomposed into a spatially uniform thermodynamic pressure $P_{th}(t)$ and a local dynamic/hydrostatic part $p^*(\mathbf{x}, t)$:

$$P(\mathbf{x}, t) = P_{th}(t) + p^*(\mathbf{x}, t) = P_{th}(t) + \bar{p}(\mathbf{x}, t) + \rho_0 g_i x_i \tag{1}$$

The complete set of Favre-filtered low-Mach conservation equations in Cartesian coordinates $(x_i) = (x, y, z)$ is given by [3]:

$$\frac{\partial \bar{\rho}}{\partial t} + \frac{\partial}{\partial x_j}(\bar{\rho}\tilde{u}_j) = 0 \tag{2}$$

$$\frac{\partial(\bar{\rho}\tilde{u}_i)}{\partial t} + \frac{\partial}{\partial x_j}(\bar{\rho}\tilde{u}_j\tilde{u}_i) = -\frac{\partial \bar{p}}{\partial x_i} + \frac{\partial}{\partial x_j}(\tau_{ij} + \tau_{ij}^{sgs}) + (\bar{\rho} - \rho_0)g_i \tag{3}$$

$$\frac{\partial(\bar{\rho}\tilde{h})}{\partial t} + \frac{\partial}{\partial x_j}(\bar{\rho}\tilde{u}_j\tilde{h}) = \frac{\partial}{\partial x_j}(\phi_{c,j} + \phi_{r,j} + \phi_{s,j}) + \frac{dP_{th}}{dt} \tag{4}$$

$$\frac{\partial(\bar{\rho}\tilde{Y}_\alpha)}{\partial t} + \frac{\partial}{\partial x_j}(\bar{\rho}\tilde{u}_j\tilde{Y}_\alpha) = \frac{\partial}{\partial x_j}(j_{m,\alpha,j} + j_{s,\alpha,j}) + \tilde{\omega}_\alpha \tag{5}$$

where $\bar{\rho}$ is the filtered density, $\tilde{u}_i$ is the Favre-filtered velocity component, $\tilde{h}$ is the Favre-filtered enthalpy, $\tilde{Y}_\alpha$ is the Favre-filtered mass fraction of chemical species α, and $\bar{p}$ is the Reynolds-filtered dynamic pressure [3].

Because the fluid undergoes volumetric expansion ($\frac{\partial \tilde{u}_i}{\partial x_i} \neq 0$), the viscous stress tensor τ_{ij} incorporates dilatation effects:

$$\tau_{ij} = \mu \left(\frac{\partial \tilde{u}_i}{\partial x_j} + \frac{\partial \tilde{u}_j}{\partial x_i} - \frac{2}{3} \frac{\partial \tilde{u}_k}{\partial x_k} \delta_{ij} \right) \tag{6}$$

where μ is the dynamic viscosity. The fluid system is closed via the Ideal Gas equation of state for the mixture:

$$P_{th}(t) = \bar{\rho} R_g \tilde{T}, \quad R_g = \frac{\mathcal{R}}{W}, \quad \frac{1}{W} = \sum_\alpha \frac{\tilde{Y}_\alpha}{W_\alpha} \tag{7}$$

where $\mathcal{R} = 8.314$ J/(mol · K) is the universal gas constant, $\tilde{T}$ is the filtered temperature, and W_α are the molar weights [3].

4.1.1 The Boussinesq Approximation Regime for Baseline Validation

For conditions where temperature variations across vertical walls are sufficiently small, fluid density changes can be neglected everywhere except in the buoyancy term. Under this assumption, density is approximated via a first-order Taylor expansion:

$$\rho = \rho_0 \left(1 - \beta (\tilde{T} - T_0) \right) \tag{8}$$

where β is the thermal expansion coefficient ($\beta = -\frac{1}{\rho_0} \left. \frac{\partial \rho}{\partial T} \right|_p$). For small temperature differences ($\beta(\tilde{T} - T_0) \ll 1$), density fluctuations vanish from transient and convective terms, simplifying the continuity and momentum equations to:

$$\frac{\partial \tilde{u}_i}{\partial x_i} = 0 \tag{9}$$

$$\frac{\partial (\rho_0 \tilde{u}_i)}{\partial t} + \frac{\partial}{\partial x_j} (\rho_0 \tilde{u}_j \tilde{u}_i) = -\frac{\partial \bar{p}}{\partial x_i} + \frac{\partial \tau_{ij}}{\partial x_j} - \rho_0 \beta (\tilde{T} - T_0) g_i \tag{10}$$

where $\frac{\partial \tau_{ij}}{\partial x_j}$ represents the simplified viscous stress divergence derived consistently from the generalized expression in Eq. (6) under the divergence-free velocity constraint $\left(\frac{\partial \tilde{u}_i}{\partial x_i} = 0 \right)$.

4.1.2 Thermal Conduction and Species Mass Diffusion

In the enthalpy conservation equation, the total enthalpy $\tilde{h}$ is expressed as a mass-weighted average of species enthalpies:

$$\tilde{h} = \sum_\alpha \tilde{Y}_\alpha h_\alpha(\tilde{T}), \quad h_\alpha(\tilde{T}) = \int_{T_0}^{\tilde{T}} c_{p,\alpha}(\theta) d\theta + \Delta h^0_{f,\alpha} \tag{11}$$

where $T_0 = 298.15$ K, $c_{p,\alpha}$ is the specific heat capacity, and $\Delta h^0_{f,\alpha}$ is the enthalpy of formation. The conductive thermal flux vector $\phi_{c,j}$, molecular mass flux vector $j_{m,\alpha,j}$, and their respective subgrid-scale (SGS) turbulent fluxes $\phi_{s,j}$ and $j_{s,\alpha,j}$ are modeled using gradient diffusion formulations:

$$\phi_{c,j} = -\frac{\mu}{Pr} \frac{\partial \tilde{h}}{\partial x_j}, \quad \phi_{s,j} = -\frac{\mu_t}{Pr_t} \frac{\partial \tilde{h}}{\partial x_j} \tag{12}$$

$$j_{m,\alpha,j} = -\frac{\mu}{\bar{\rho} Sc} \frac{\partial \tilde{Y}_\alpha}{\partial x_j}, \quad j_{s,\alpha,j} = -\frac{\mu_t}{\bar{\rho} Sc_t} \frac{\partial \tilde{Y}_\alpha}{\partial x_j} \tag{13}$$

with constant turbulent Prandtl and Schmidt numbers ($Pr_t = Sc_t = 0.7$) and turbulent dynamic viscosity μ_t.

4.1.3 Radiative Transfer Equation (RTE) and Coupling

Radiative heat losses and thermal feedback divergence ($\frac{\partial \phi_{r,j}}{\partial x_j}$) are accounted for by solving the non-spectral Radiative Transfer Equation (RTE) along a specific line-of-sight direction vector $\mathbf{s}$ for an absorbing-emitting medium with intensity $I(\mathbf{x}, \mathbf{s})$ [3]:

$$s_j \frac{\partial I(\mathbf{x}, \mathbf{s})}{\partial x_j} + \kappa I(\mathbf{x}, \mathbf{s}) = \kappa I_b, \quad I_b = \frac{\sigma \tilde{T}^4}{\pi} \tag{14}$$

where σ is the Stefan-Boltzmann constant, I_b is blackbody intensity, and κ is the total absorption coefficient. The coupling between radiation and the thermal energy equation is established through the divergence of the net radiative heat flux vector component $\phi_{r,j} = \int_{4\pi} I(\mathbf{x}, \mathbf{s}) s_j \, d\Omega$, expressed as:

$$-\frac{\partial \phi_{r,j}}{\partial x_j} = \kappa \left(G - 4\sigma \tilde{T}^4 \right), \quad G = \int_{4\pi} I(\mathbf{x}, \mathbf{s}) \, d\Omega \tag{15}$$

where G represents the incident radiation integrated over all solid angles 4π. In this work, the total absorption coefficient κ comprises:

1. Gaseous absorption (κ_g) calculated via the Weighted Sum of Grey Gases (WSGG) model for combustion products (H_2O and CO_2) [3].
2. Soot absorption (κ_s) evaluated using soot mass fractions to account for participating medium radiation [3].

4.1.4 Combustion Modelling

To simulate the wall-fired compartment scenarios using a single gaseous fuel, a single-step global Eddy-Dissipation Model (EDM) implemented within the Large Eddy Simulation framework of CALIF^{3}S is utilized [3]. The global combustion reaction pathway is expressed as:

$$F + sO_2 \rightarrow \sum_i s_{p,i} P_i \tag{16}$$

where F represents the fuel, O_2 is the oxidizer, s is the stoichiometric mass ratio of oxygen to fuel, and $s_{p,i}$ are the stoichiometric coefficients of the combustion products P_i (H_2O, CO_2, and soot).

In accordance with the single-fuel EDM formulation, the chemical reaction rate is assumed to be infinitely fast compared to fluid mixing scales, meaning combustion is entirely controlled by turbulent mixing [3]. Following a conserved scalar framework analogous to the general scalar transport structure of Eq. (5), linear combinations of species mass fractions yield Shvab-Zeldovich conserved scalar variables β_i (such as $\beta_0 = s\tilde{Y}_F - \tilde{Y}_{O_2}$). The normalized mixture fraction Z is subsequently defined as:

$$Z = \frac{\beta_i - \beta_i^o}{\beta_i^f - \beta_i^o} \tag{17}$$

where the boundary stream parameters are explicitly defined as:

- β_i^f represents the conserved scalar value evaluated at the pure fuel stream boundary condition (i.e. $Z = 1$).
- β_i^o represents the conserved scalar value evaluated at the pure oxidizer stream boundary condition (i.e. $Z = 0$).

The transport equations solved for the mean mixture fraction $\tilde{Z}$ and the mean fuel mass fraction $\tilde{Y}_F$ are given by [3]:

$$\frac{\partial}{\partial t}(\bar{\rho}\tilde{Z}) + \frac{\partial}{\partial x_j}(\bar{\rho}\tilde{u}_j\tilde{Z}) = \frac{\partial}{\partial x_j}(j_{m,Z,j} + j_{s,Z,j}) \tag{18}$$

$$\frac{\partial}{\partial t}(\bar{\rho}\tilde{Y}_F) + \frac{\partial}{\partial x_j}(\bar{\rho}\tilde{u}_j\tilde{Y}_F) = \frac{\partial}{\partial x_j}(j_{m,F,j} + j_{s,F,j}) + \tilde{\omega}_F \tag{19}$$

where $j_{m,Z,j}, j_{s,Z,j}$ and $j_{m,F,j}, j_{s,F,j}$ are the molecular and subgrid-scale diffusion fluxes for the mixture fraction and fuel mass fraction, respectively, and $\tilde{\omega}_F$ is the mean fuel consumption rate. As with the general species scalar $\tilde{Y}_\alpha$ in Eq. (5), these fluxes follow the same gradient-diffusion closures introduced in Eqs. (13)-(14), with the generic species index α replaced by Z and F respectively:

$$j_{m,Z,j} = -\frac{\mu}{\bar{\rho}Sc}\frac{\partial\tilde{Z}}{\partial x_j}, \quad j_{s,Z,j} = -\frac{\mu_t}{\bar{\rho}Sc_t}\frac{\partial\tilde{Z}}{\partial x_j} \tag{20}$$

$$j_{m,F,j} = -\frac{\mu}{\bar{\rho}Sc}\frac{\partial\tilde{Y}_F}{\partial x_j}, \quad j_{s,F,j} = -\frac{\mu_t}{\bar{\rho}Sc_t}\frac{\partial\tilde{Y}_F}{\partial x_j} \tag{21}$$

with the same constant Schmidt numbers (Sc, $Sc_t = 0.7$) as defined in Section 4.1.2.

Within the LES framework, the fuel burning rate $\tilde{\omega}_F$ is closed using the subgrid characteristic time scale τ [3]:

$$\tilde{\omega}_F = -C_R\bar{\rho}\frac{1}{\tau}\min\left(\tilde{Y}_F, \frac{\tilde{Y}_{O_2}}{s}\right) \tag{22}$$

where $C_R = 1$ is the default dimensionless reaction constant, $\tilde{Y}_{O_2}$ is the filtered oxygen mass fraction, and the inverse time scale for LES is modeled via the subgrid diffusion coefficient D_{LES} and filter width Δ_{LES} as [3]:

$$\frac{1}{\tau} = \frac{D_{LES}}{C_{LES}\Delta_{LES}^2} \tag{23}$$

Here, $C_{LES} = 0.1$ is the default subgrid model constant. Combining these coefficients ($C_R/C_{LES} = 1/0.1$) yields an effective proportionality factor of 10 implemented in the solver configuration, directly coupling local subgrid turbulent mixing to the heat release rate of the buoyant fire plume.

4.1.5 Soot Modelling

Predicting transient soot formation via detailed aerosol kinetics is computationally prohibitive for large-scale Large Eddy Simulations. Therefore, following standard fire engineering practices and the built-in capabilities of CALIF^{3}S, soot generation is accounted for using the "Simple Approach" soot model [3].

In this framework, instead of solving a separate transport equation for soot mass fraction, soot production is integrated directly into the single-step global combustion reaction of a generalized hydrocarbon fuel (C_nH_m) via a constant soot conversion factor (stoichiometric coefficient ν_s) [3]:

$$C_nH_m + \left(n + \frac{m}{4} - \nu_s\right)O_2 \rightarrow (n - \nu_s)CO_2 + \frac{m}{2}H_2O + \nu_sC \tag{24}$$

The local soot mass fraction $\tilde{Y}_s$ is then dynamically evaluated from the mixture fraction and fuel mass fractions using the pure fuel stream mass fraction ($\tilde{Y}_F^f$) as [3]:

$$\tilde{Y}_s = \frac{\nu_s W_C}{\nu_F W_F} \left(\tilde{Y}_F^f - \tilde{Y}_F \right) \tag{25}$$

where W_C and W_F are the molar masses of carbon and fuel, respectively, and $\tilde{Y}_F$ is the filtered fuel mass fraction[cite: 3]. This computed soot distribution couples directly with the non-spectral WSGG radiation model to determine the total absorption coefficient ($\kappa = \kappa_g + \kappa_s$) throughout the domain, ensuring accurate radiative heat feedback to the boundary walls[3].

4.2 Subgrid Scale Models

In Large Eddy Simulation (LES), resolving the entire spectrum of turbulent scales is computationally prohibitive. Subgrid-scale (SGS) models are critically important to achieve numerical closure by accounting for the dissipative effects of the small, unresolved eddies on the large, resolved structures. Following the Boussinesq assumption as defined in the CALIF^{3}S Physical Models Framework [3], the filtered subgrid-scale stress tensor (τ_{ij}^{sgs}) is modeled via a subgrid eddy viscosity μ_s:

$$\tau_{ij}^{sgs} - \frac{2}{3} k_{sgs} \delta_{ij} = -2\mu_s \bar{S}_{ij} \tag{26}$$

Where $\bar{S}_{ij} = \frac{1}{2}\left(\frac{\partial \tilde{u}_i}{\partial x_j} + \frac{\partial \tilde{u}_j}{\partial x_i} \right)$ represents the filtered rate-of-strain tensor in Cartesian coordinates [3].

To achieve closure for the unclosed terms arising from the Favre-filtering process in Equations (3) and (4), subgrid-scale models are introduced. These models account for the dissipative effects of unresolved subgrid eddies on the large, resolved flow structures by modeling the subgrid stress tensor (τ_{ij}^{sgs}), the subgrid enthalpy/heat flux ($\phi_{s,j}$), and the subgrid scalar species fluxes ($j_{s,\alpha,j}$).

4.2.1 Limitations of Classical Approaches

The classical Smagorinsky model defines subgrid eddy viscosity as:

$$\mu_s = \bar{\rho}(C_s \Delta)^2 |\bar{S}| \tag{27}$$

Where C_s is a model parameter, Δ is the characteristic cell size, and $|\bar{S}| = \left(2\bar{S}_{ij}\bar{S}_{ij}\right)^{1/2}$ [3]. While functional for isotropic turbulence, this formulation introduces severe unphysical behavior in wall-bounded scenarios by yielding a non-zero eddy viscosity directly at solid physical boundaries [3]. Overcoming this requires computationally intrusive, ad-hoc damping corrections (e.g., Van Driest functions) which fail in complex or transitional buoyant flows.

4.2.2 The Wall-Adapting Local Eddy-viscosity (WALE) Model

To resolve these near-wall discrepancies without the numerical instabilities typical of dynamic sampling procedures, the WALE model proposed by Nicoud and Ducros (1999) [4] is integrated. The model formulates the subgrid eddy viscosity based on the square of the velocity gradient tensor [3]:

$$\mu_s = \bar{\rho}(C_w \Delta)^2 \frac{\left(\sum_{i,j} \tilde{g}_{ij} \tilde{g}_{ij} \right)^{3/2}}{\left(\sum_{i,j} \bar{S}_{ij} \bar{S}_{ij} \right)^{5/2} + \left(\sum_{i,j} \tilde{g}_{ij} \tilde{g}_{ij} \right)^{5/4}} \tag{28}$$

Where C_w is the designated WALE model constant (set to the default validation value of 0.5 in CALIF^{3}S) [3, 4], and $\tilde{g}_{ij}$ represents the traceless symmetric part of the squared velocity gradient tensor, formulated in index notation as [3]:

$$\tilde{g}_{ij} = \frac{1}{2}\left(g_{ik}g_{kj} + g_{jk}g_{ki}\right) - \frac{1}{3}\delta_{ij}g_{mk}g_{km}, \quad g_{ij} = \frac{\partial \tilde{u}_i}{\partial x_j} \tag{29}$$

The mathematical architecture of the WALE model provides two primary physical advantages for this vertical channel configuration [4]:

1. **Asymptotic Near-Wall Scaling:** The formulation naturally reproduces the correct physical scaling of eddy viscosity near solid boundaries ($\mu_s \propto y^3$, where y is the normal distance to the wall) [3, 4]. It automatically goes to zero at the non-slip boundaries without requiring empirical damping functions [3].
2. **Sensitivity to Shear vs. Rotation:** By accounting for both strain and rotation rates, the model does not produce spurious subgrid viscosity in pure shear flow zones [4]. This allows natural laminar-to-turbulent boundary transitions to emerge organically.

4.3 Numerical Schemes

To preserve the chaotic, energy-containing turbulent fluctuations resolved in a Large Eddy Simulation (LES) framework, low-dissipation and high-order accurate numerical discretization schemes must be utilized. Standard upwind approximations introduce excessive numerical diffusion that artificially damps out small-scale subgrid motions. Within the CALIF^{3}S software environment, the spatial and temporal operators are discretized using the following configurations:

4.3.1 Spatial Discretization Schemes

To preserve the chaotic, energy-containing turbulent fluctuations resolved in a Large Eddy Simulation (LES) framework, low-dissipation and high-order accurate numerical discretization schemes must be utilized. Standard upwind approximations introduce excessive numerical diffusion that artificially damps out small-scale subgrid motions. Within the CALIF^{3}S software environment, the spatial and temporal operators are discretized using the following configurations:

- **Convective operator of the NS equations:**
  - Kinetic energy preserving 2nd-order centered scheme.
  - 3rd-order Quick with deferred upwind correction
  - 3rd-order Quick, explicit.
- **Convective operator of the scalar transport equations:**
  - 3rd-order Quick, explicit.
  - 3rd-order Quick with deferred implicit correction (semi-quick implicit upwind).
- **Diffusive / Viscous Terms:** Discretized utilizing a standard **2nd-order Centered Finite-Volume Scheme**. This guarantees precise estimation of the steep molecular shear gradients and conductive heat transfer rates directly at the solid wall boundaries.

4.3.2 Temporal Discretization (Time-Stepping)

To capture the highly unsteady, time-varying nature of natural convection plumes, an accurate time-integration method is mandatory:

- **Temporal Advancement:** Discretized using a **2nd-order Implicit Scheme** (Dual Time-Stepping formulation). This ensures that temporal accuracy scales quadratically as $\mathcal{O}(\Delta t^2)$, providing a strong mathematical balance between structural stability and transient tracking.
- **Time-Step and CFL Constraints:** The time-advancement interval (Δt) is dynamically or statically regulated to satisfy the Courant-Friedrichs-Lewy (CFL) stability criterion:

$$CFL = \frac{||\mathbf{v}|| \cdot \Delta t}{\Delta x} \leq Co_{\max} \tag{30}$$

  For the high-temperature variable-density cases, this parameter is systematically tightened to prevent solver divergence caused by massive velocity spikes.

4.4 Criteria for LES Convergence and Mesh Quality Assessment

To verify the fidelity of the Large Eddy Simulation (LES) within the differentially heated vertical channel, a dual-zone quality assessment framework is implemented. Following the operational parameters of the solver, the criteria are divided into two distinct physical regimes: the *Parietal Zone Criteria* governing near-wall resolution and the *Free-Stream Criteria* governing the core isotropic turbulent scales.

4.4.1 Parietal Zone Criteria (Near-Wall Region)

The parietal zone criteria track the statistical convergence and spatial resolution of the thin viscous and thermal boundary layers adhering to the channel walls. To ensure that the grid spacing resolves the viscous sublayer without relying on empirical wall functions as $\delta w \to 0$, the following conditions must be fulfilled:

1. **Dimensionless Wall Distance (y^+):** The spatial node closest to the solid boundary must satisfy:

$$y^+ \leq 1 \tag{31}$$

2. **Skin Friction Coefficient (C_f):** The hydrodynamic boundary layer convergence is tracked via the time-averaged skin friction coefficient, defined using a buoyant reference velocity scale u_e:

$$C_f = \frac{\mu \frac{\delta w}{\delta y}}{\frac{1}{2}\rho_e u_e^2}, \quad \text{with} \quad u_e = \sqrt{\frac{\Delta T \cdot g \cdot h}{T}} \tag{32}$$

   Where $\delta w/\delta y$ is the local near-wall velocity gradient, h is the channel height, and ρ_e is the characteristic density.

3. **Nusselt Number (Nu):** The thermal boundary layer development and wall heat flux stability are assessed using the non-dimensional temperature gradient:

$$Nu = \frac{L \frac{\delta T}{\delta y}}{\Delta T} \tag{33}$$

Statistical temporal convergence is achieved when the running averages of both C_f and Nu achieve a stable, invariant plateau over time.

4.4.2 Free-Stream Criteria (Channel Core Region)

Within the central core of the vertical channel, the grid must be fine enough to resolve the primary energy-containing turbulent fluctuations. The spatial resolution capacity of the local grid size (Δ) is evaluated using Pope and Celik structural metrics:

1. **Pope Energy Resolution Index:** The ratio of the resolved turbulent kinetic energy ($k_{t,R}$) to the total turbulent kinetic energy (the sum of resolved and subgrid-scale kinetic energy, $k_{t,R} + k_{t,S}$) must demonstrate that the mesh directly captures at least 95% of the flow's total energy:

$$\text{Pope:} \quad \frac{k_{t,R}}{k_{t,R} + k_{t,S}} \geq 0.95 \tag{34}$$

Where the resolved turbulent kinetic energy is defined as:

$$k_{t,R} = \frac{1}{2}(\tilde{u}_i - \langle\tilde{u}_i\rangle)(\tilde{u}_i - \langle\tilde{u}_i\rangle) \tag{35}$$

and the subgrid-scale kinetic energy is modeled as:

$$k_{t,S} = C_K \left(\frac{\epsilon_S \Delta}{\pi}\right)^{2/3} \quad \text{with} \quad C_K = 1.5 \tag{36}$$

where $\langle\tilde{u}_i\rangle$ denotes the time-averaged (or ensemble-averaged) component of the Favre-filtered velocity, utilized to isolate the resolved turbulent fluctuations from the mean flow.

2. **Celik Grid Resolution Index:** To ensure that the filter width of the computational mesh (Δ) approaches the dissipative structures of the flow, the grid-to-Kolmogorov length scale ratio must satisfy:

$$\text{Celik:} \quad \frac{\Delta}{\eta_K} \leq 25 \tag{37}$$

Where the Kolmogorov microscale η_K is derived from the molecular kinematic viscosity ν and the total energy dissipation rate ϵ (comprising both resolved dissipation ϵ_R and subgrid-scale dissipation ϵ_S):

$$\eta_K = \left(\frac{\nu^3}{\epsilon}\right)^{1/4}, \quad \text{with} \quad \epsilon = \epsilon_R + \epsilon_S \tag{38}$$

with the respective dissipation components expressed as:

$$\epsilon_R = \left\langle \frac{2\mu_e}{\bar{\rho}} \tilde{s}_{ij}\tilde{s}_{ij} \right\rangle, \quad \epsilon_S = \left\langle \frac{2\mu_S}{\bar{\rho}} \tilde{s}_{ij}\tilde{s}_{ij} \right\rangle \tag{39}$$

where $\langle \dots \rangle$ denotes the statistical time-averaging operator applied to the local dissipation components to obtain mean dissipation rates.

5 LES of Differentially Heated Vertical Channel

5.1 General Description

The geometry consists of a rectangular channel of total width $H = 0.1$ m, where the wall-normal distance is defined as $y/H \in [0, 1]$.

5.1.1 Domain Dimensions

The computational domain extends beyond the target height (h) to ensure fully developed turbulence:

- **Streamwise length (L_z):** $1.2\,\mathrm{m}$ (Periodic Boundary Condition)
- **Spanwise width (L_x):** $0.6\,\mathrm{m}$ (Periodic Boundary Condition)
- **Wall-normal distance ($L_y = H$):** $0.1\,\mathrm{m}$ (Dirichlet Thermal Boundary Condition)

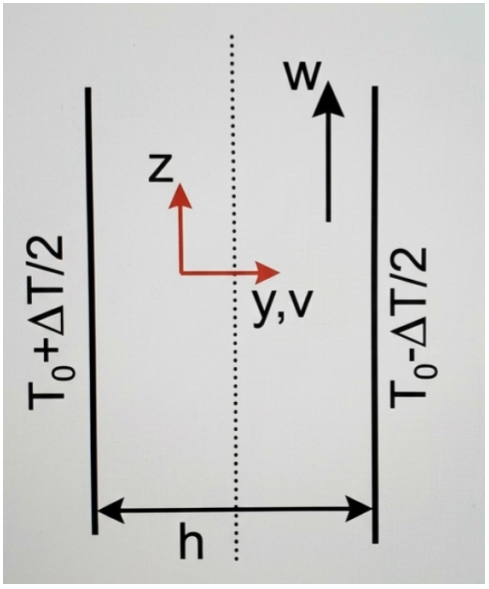


Figure 2. Schematic of the Differentially Heated Vertical Channel domain. x, z directions are periodic; y is wall-normal.

5.1.2 Mesh Architecture

A structured mesh of resolution N_x (Span) $\times$ N_y (Wall-Normal) $\times$ N_z (Stream) nodes was generated. A mixed meshing strategy was employed:

1. **Periodic Axes (Span and Stream):** Defined by a *regular vector*, ensuring uniform spacing.
2. **Wall-Normal Axis (L_y):** Defined by a *stretched vector* formulation, ensuring a fine grid resolution near the vertical plates to resolve the viscous and thermal sublayers, with coarser resolution towards the channel centerline.

The localized wall-adjacent cell height (Δy_{wall}) closest to the solid physical boundaries is calculated using the core reference spacing (Δ) and an empirical wall stretching factor (r_{wall}) via:

$$\Delta y_{\text{wall}} = \Delta \cdot r_{\text{wall}} \tag{40}$$

5.1.3 Temporal Discretization

The simulation utilized a time step adapted based on the delta(max) value in the wall normal direction, calculated directly from the CFL formula to ensure numerical stability. The total simulation duration was:

- **Statistical Restarting Time (t_{start}):** $50\,\mathrm{s}$ Initialized start time to capture a developed field
- **End Time (t_{end}):** $100\,\mathrm{s}$

This provides a significant time window for statistical averaging.

5.1.4 Fluid Properties and Parameter Adaptation

To match $Ra = 5.4 \times 10^5$, non-dimensional consistency was maintained with the reference DNS (Prandtl number $Pr \approx 0.71$). This configuration serves as the foundational simulation setup in this work, designed primarily to establish baseline code validation against high-fidelity reference Direct Numerical Simulation (DNS) database papers. By evaluating the system at a well-documented Rayleigh number, the predictive capabilities of the Large Eddy Simulation (LES) approach and the core implementation properties of the solver can be rigorously verified.

The physical properties and operational parameters implemented in the solver were:

Table 1. Simulation Parameters and Fluid Properties for Rayleigh 5.4×10^5

Parameter	Symbol / Value	Unit
Thermal Diffusivity	$\alpha = 6.47 \times 10^{-5}$	$\mathrm{m^2\,s^{-1}}$
Kinematic Viscosity	$\nu = 4.53 \times 10^{-5}$	$\mathrm{m^2\,s^{-1}}$
Gravity	$g = 9.81$	$\mathrm{m\,s^{-2}}$
Prandtl Number	$Pr = 0.71$	–
Hot Wall Temperature	$T_{\text{hot}} = 324.26$	K
Cold Wall Temperature	$T_{\text{cold}} = 275.74$	K
Temperature Difference	$\Delta T = 48.5$	K
Channel Width	$H = 0.1$	m

To ensure that the numerical solutions are independent of the spatial discretization parameters comprehensive sensitivity studies were executed across a systematically varied grid matrix.

5.2 Averaging Procedure

To eliminate this spatial sampling bias, the post-processing methodology was modified to utilize the entire 3D volume dataset exported from the solver into ParaView. Rather than relying on a line probe, spatial averaging was executed over the full homogeneous plane coordinates according to the following mathematical formulation:

$$\langle K \rangle(y) = \frac{1}{L_x L_z} \int_0^{L_x} \int_0^{L_z} K(x, y, z, t)\, dx\, dz \tag{41}$$

where $\langle\rangle$ denotes the spatial plane-averaging operator over the homogeneous x and z coordinates for statistical convergence

By extracting the complete volume dataset and carrying out plane-integration across the periodic boundaries, the localized non-invariances were completely smoothed out. This methodological shift successfully recovered the balanced, symmetrical double-peak TKE profiles characteristic of fully developed buoyancy-driven turbulence, ensuring strict validation against the Versteegh DNS reference data.

5.3 Description of the Simulated Cases $Rayleigh = 5.4 \times 10^5$ Case

The complete dataset tracking critical boundary layer and statistical convergence metrics across the eleven simulated cases is compiled in Table 2. The tracked engineering metrics include the non-dimensional wall distance (y^+), the time-converged Nusselt number (Nu), the skin friction

coefficient (C_f), the Pope turbulence resolution criterion, and the Celik grid refinement index.The schemes are listed in the order of Scalar convective and Momentum convective (Navier-stokes) schemes

Table 2. Sensitivity analysis matrix for the Boussinesq-coupled channel flow simulation series (Simulation 1 serves as the baseline reference configuration).

Sim.	Δs	r_{wall}	Hydro.	LES Model	Scheme (Con-N/S)	y^+	Nu	C_f	Celik
1	0.012	0.1	Bous.	WALE	QUICK-Cent.	1.1631	5.0007	3.039e-2	7.604
2	0.024	0.05	Bous.	WALE	QUICK-Cent.	1.1567	4.4422	4.747e-2	14.90
3	0.006	0.2	Bous.	WALE	QUICK-Cent.	0.9642	5.3436	2.704e-2	4.022
4	0.012	0.2	Bous.	WALE	QUICK-Cent.	1.7630	4.1675	1.811e-2	8.667
5	0.012	0.5	Bous.	WALE	QUICK-Cent.	1.5430	1.8815	3.477e-3	10.04
6	0.012	1	Bous.	WALE	QUICK-Cent.	3.1984	0.6988	2.718e-2	11.05
7	0.012	0.1	Bous.	WALE	QUICK-QUICK	1.2100	4.2296	3.289e-2	7.346
8	0.012	0.1	Bous.	WALE	QUICK-Imp.	1.2795	4.7570	3.678e-2	7.354
9	0.012	0.1	Bous.	Smagorinsky	QUICK-Cent.	1.7796	0.9430	7.114e-2	13.14
10	0.012	0.1	Bous.	D-Smago.	QUICK-Cent.	1.1390	5.7274	2.917e-2	7.721

5.3.1 Reference Simulation

To establish a comprehensive validation framework, the localized time-averaged profiles of Simulation 1 are evaluated directly against the established DNS benchmark data from Versteegh.This comparison maps the spatial distribution of velocity, temperature, turbulent kinetic energy (TKE), thermal dissipation, and variance tracking across the normalized channel width (y/H).

5.3.2 Mean Field Symmetry and Buoyancy Coupling

As illustrated in Figure 3, the mean vertical velocity (Wj) and temperature (T^*) profiles exhibit excellent antisymmetric and symmetric spatial profiles, respectively. These non-dimensional variables are defined as:

$$W^* = \frac{V \cdot h}{\alpha}, \quad \text{with} \quad \alpha = \frac{\nu}{Pr} \tag{42}$$

and

$$T^* = \frac{T - T_{\text{avg}}}{\Delta T} \tag{43}$$

- **Velocity Field (W^*):** The velocity curve clearly captures the buoyancy-driven up-flow zone adjacent to the heated boundary wall ($y/H \to 0$) and the matching down-flow zone near the cooling wall ($y/H \to 1$) . The peak velocities and local shear layers align tightly with the DNS benchmarks, indicating that the driving buoyancy source vectors are correctly balanced .
- **Thermal Field (T^*):** The temperature profile traces a steep gradient through the linear thermal sublayers at both walls, transitioning into a uniform, well-mixed core region across the center of the channel ($0.2 < y/H < 0.8$) . This highlights that the Boussinesq model accurately maintains spatial thermal equilibrium .

5.3.3 Turbulent Kinetic Energy and Spectral Partitioning

Figure 4 evaluates the distribution of the Total Turbulent Kinetic Energy (K_{total}) alongside its spectral components .

To maintain non-dimensional consistency with reference data, the quantities are scaled using specific framework controls. For instance, the non-dimensional Turbulent Kinetic Energy (TKE*)

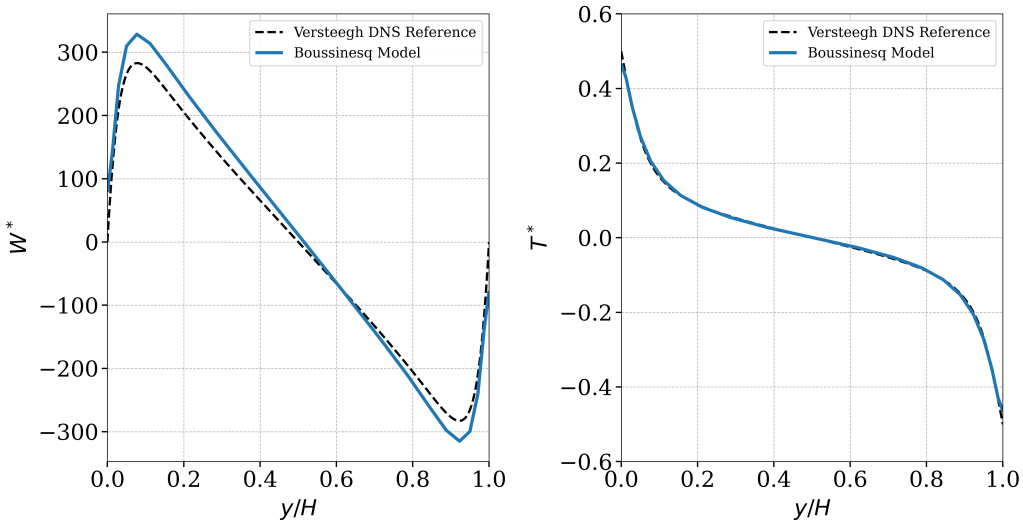


Figure 3. Comparison of mean vertical velocity W^* and non-dimensional temperature T^* profiles against DNS reference curves.

is scaled via the following relationship:

$$K^*_{\text{total}} = \text{TKE} \cdot \frac{h_c^2}{\alpha^2 \cdot S_{\text{paper}}} \tag{44}$$

where h_c represents the half-channel width, α is the thermal diffusivity, and $S_{\text{paper}} = Ra^{8/9}$ denotes the paper scaling factor corresponding to the Rayleigh number.

To properly interpret the behavior observed in the turbulent kinetic energy (TKE) and dissipation profiles, it is essential to recognize the underlying physical cascade unique to closed differentially heated cavities. Unlike forced flows where momentum is externally injected, the flow dynamics here follow a strict thermal-to-mechanical chain: the wall-to-fluid temperature gradient (ΔT) acts as the primary driver, inducing density differences that generate vertical buoyant velocities (U). This velocity development subsequently creates sharp spatial velocity gradients ($\partial U/\partial y$) near the walls, which dictate the local shear, turbulent kinetic energy production, and ultimately the subgrid-scale profiles.

- **DNS Agreement:** The resolved WALE structure tracks the structural profile of the reference budget curve across the entire channel, slightly over-predicting the peak values in the high-shear region near the walls before settling into the core value .
- **Resolution Scale Evaluation:** The bottom plot reveals that the resolved fluid scale captures the vast majority of the total energy budget across the domain . The subgrid-scale contribution remains consistently low, satisfying the standard requirement that an acceptable LES mesh must directly resolve more than 80–90% of the total turbulent kinetic energy.

5.3.4 Dissipation and Fluctuating Fields

The fluctuating thermal structures are tracked via the local thermal dissipation (ε^*_θ) and normalized temperature variance ($\overline{T'^2}/\Delta T^2$) fields in Figures 5 and 6 .

- **Thermal Dissipation (ϵ^*_θ):** Scaled via $\dfrac{h^2}{\alpha \cdot \Delta T^2}$, where half-height $h = 0.05$, $\alpha = 6.47\times10^{-5}$, and $\Delta T = 48.52$.

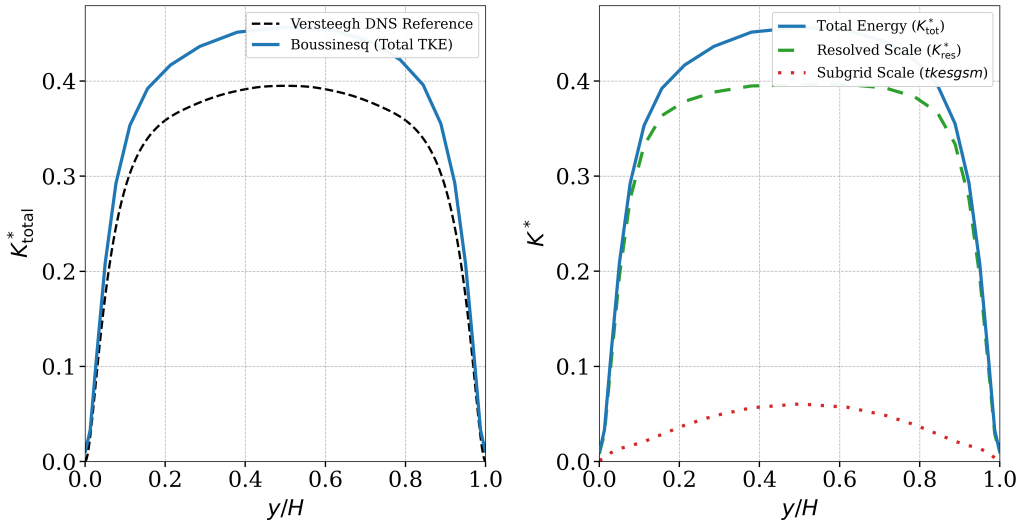


Figure 4. Total Turbulent Kinetic Energy (K_{total}) distribution and spectral partitioning breakdown

- **Temperature Variance ($T'T'$):** Scaled via $\dfrac{1}{\Delta T^2}$, where the temperature difference $\Delta T = 48.52$.
- **Thermal Dissipation Profile (ε^*_θ):** The dissipation rate peaks sharply in the immediate vicinity of the walls due to high local thermal gradients before dropping to a low, stable value in the channel core . The spectral partitioning shows that subgrid-scale approximations are active primarily within the near-wall buffer zone, damping out minimal small-scale energy as expected from the WALE formulation .
- **Temperature Variance ($\overline{T'^2}/\Delta T^2$):** The variance profile tracks a characteristic dual-peak distribution, with local maximums occurring near $y/H \approx 0.08$ and $y/H \approx 0.92$, matching the boundaries of the thermal buffer layer . The close agreement with DNS data confirms that the structural boundary layer behavior is well-resolved by the baseline mesh configuration .

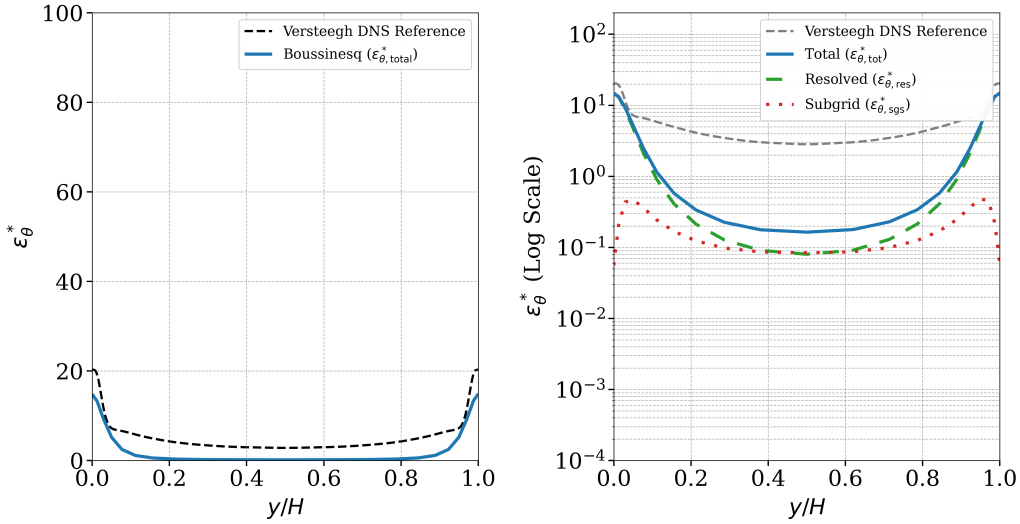


Figure 5. Thermal dissipation(ε^*_θ) and its internal resolved vs. subgrid-scale spectral split.

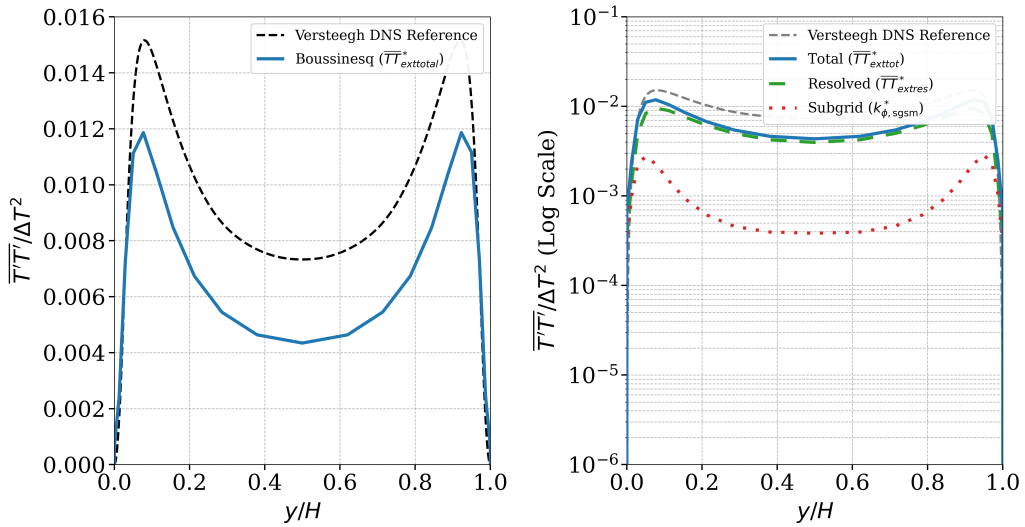


Figure 6. Temperature variance distribution profile ($\overline{T'^2}/\Delta T^2$) across the normalized domain.

5.4 Parametric Sensitivity: Influence of Core Grid Refinement (Δs)

This section evaluates the sensitivity of the Large Eddy Simulation to spatial resolution by varying the isotropic core grid spacing (Δs). The baseline configuration using a *Medium Mesh* ($\Delta s = 0.012$, Simulation 1) is compared against a *Coarse Mesh* ($\Delta s = 0.024$, Simulation 2) and a highly resolved *Fine Mesh* ($\Delta s = 0.006$, Simulation 3). Throughout this grid sensitivity analysis, the numerical scheme (QUICK-Centered), the SGS model (WALE), and the boundary wall-stretching factor ($r_{\text{wall}} = 0.1$) are held perfectly constant.

Global integrated transport metrics and validation indices for the mesh refinement cases are summarized in Table 3.

Table 3. Global transport metrics and validation criteria across different core grid resolutions.

Simulation Case	y^+	Nu	C_f	Pope Criterion	Celik Criterion
1: Medium Mesh ($\Delta s = 0.012$)	1.1631	5.0007	3.0390×10^{-2}	0.9135	7.604
2: Coarse Mesh ($\Delta s = 0.024$)	1.1567	4.4422	4.7465×10^{-2}	0.8039	14.901
3: Fine Mesh ($\Delta s = 0.006$)	0.9642	5.3436	2.7038×10^{-2}	0.9622	4.022

5.4.1 Analysis of Mean Flow Fields and Transport Behavior

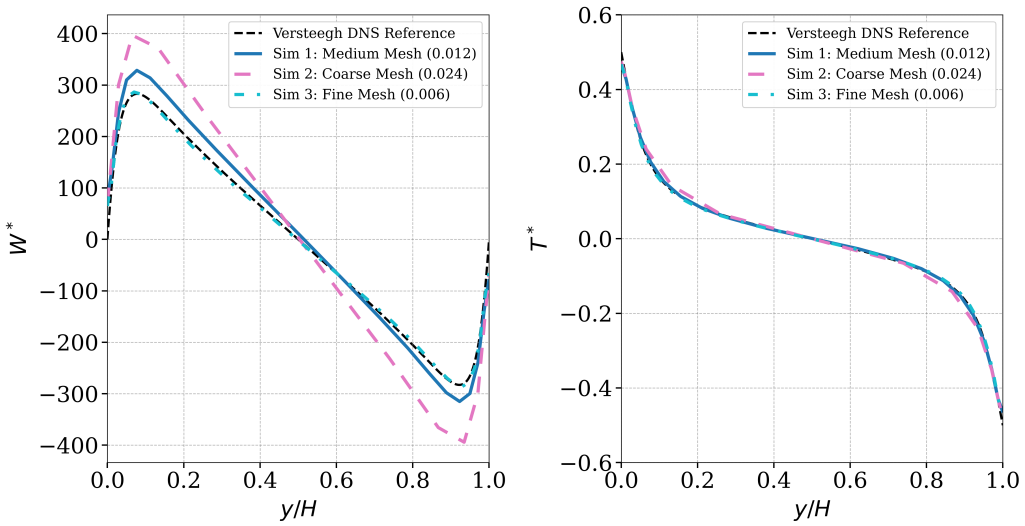


Figure 7. Comparison of mean vertical velocity profiles (W^*) and mean temperature profiles (T^*) across the normalized channel width for different core grid sizes against the DNS reference.

The mean vertical velocity (W^*) and normalized temperature (T^*) profiles across the channel width are shown in Figure 7. Looking at the velocity plot, the Coarse Mesh ($\Delta s = 0.024$) significantly overpredicts the maximum convective velocity peaks near the walls ($y/H \approx 0.05$ and 0.95). This happens because a coarse grid cannot fully resolve the sharp velocity gradients near the boundaries, primarily due to limitations in the upwind treatment of convective terms. As the grid is refined from Medium ($\Delta s = 0.012$) to Fine ($\Delta s = 0.006$), the velocity peaks systematically lower and converge toward the DNS reference curve.

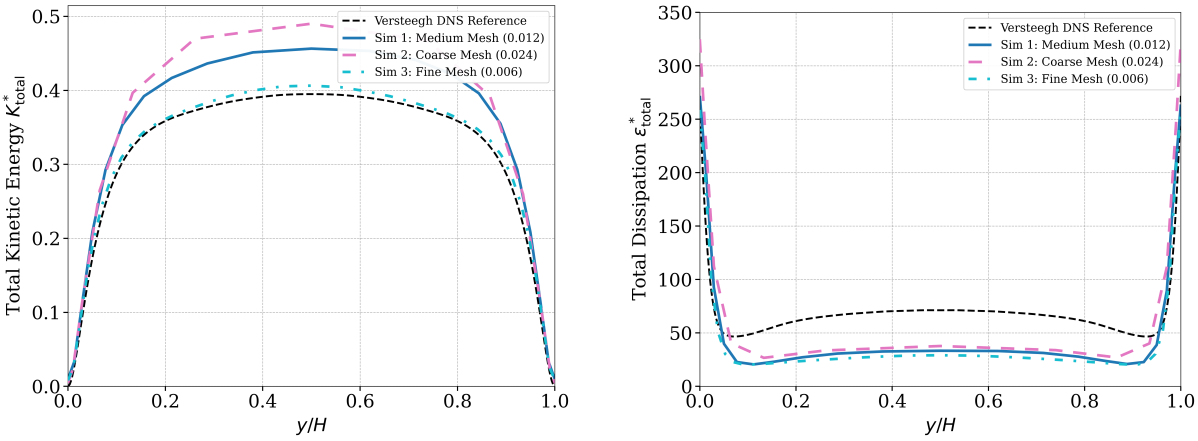
For the thermal profile (T^*), all three grid resolutions capture the overall distribution profile well. However, the Coarse Mesh exhibits a slight structural overshoot in the near-wall thermal boundary gradients. This directly influences global heat transfer calculations because near-wall temperature gradients constitute the primary source term in the Navier-Stokes momentum equations: as shown in Table 3, the Nusselt number drops to $\text{Nu} = 4.4422$ on the Coarse Mesh but converges upward to $\text{Nu} = 5.3436$ on the Fine Mesh, matching the enhanced thermal mixing seen in the reference data.

5.4.2 Turbulence Statistics and Dissipation Dynamics Across Grid Sizes

Turbulent Kinetic Energy Dynamics: The total scaled Turbulent Kinetic Energy profiles are illustrated in Figure 8a. The Coarse Mesh ($\Delta s = 0.024$) severely overpredicts the total energy budget across the entire core region of the channel ($0.2 < y/H < 0.8$). This excess energy is an artificial error: because the coarse grid cells are too large, the simulation fails to resolve small-scale turbulent structures, causing energy to build up incorrectly in the larger scales. As the grid cell size is cut in half (Medium) and quartered (Fine), this unphysical energy buildup disappears, bringing the energy distribution right down to match the true DNS profile.

Viscous Dissipation Dynamics:

The viscous dissipation ($\epsilon^*_{\text{total}}$) profiles across the channel are presented in Figure 8b. All three grid resolutions exhibit a very flat and low dissipation rate across the central core of the channel. The key difference occurs directly at the non-slip walls ($y/H = 0.0$ and 1.0). The Coarse Mesh fails to capture the massive scale of near-wall viscous friction, while the Fine Mesh captures



(a) Turbulent Kinetic Energy (K^*_{total})

(b) Viscous Dissipation ($\epsilon^*_{\text{total}}$)

(c) Thermal Dissipation (ϵ^*_θ)

(d) Temperature Variance ($\overline{T'T'}_{\text{total}}$)

Figure 8. Turbulence statistics and dissipation profiles across the normalized channel width (y/H) for different core grid sizes: (a) Total turbulent kinetic energy, (b) Viscous dissipation, (c) Total thermal dissipation, and (d) Total temperature variance.

a much larger portion of the intense near-wall dissipation spike due to its superior spatial resolution.

Thermal Dissipation Dynamics: The thermal dissipation profiles (ϵ_θ^*) are displayed in Figure 8c. Similar to the viscous dissipation curves, the profiles remain flat and uniform across the core region ($0.2 < y/H < 0.8$) for all cases. Near the solid boundaries, the Coarse Mesh underpredicts the thermal destruction rates. Refining the grid to the Fine Mesh allows the simulation to capture stronger thermal dissipation peaks at the walls, confirming that finer grid cells are needed to resolve small-scale thermal plume interactions.

Temperature Variance Profiles: The total temperature variance ($\overline{T'T'}_{\text{total}}$) profiles are plotted in Figure 8d. The Coarse Mesh suffers from a significant underprediction of the near-wall thermal fluctuation peaks ($y/H \approx 0.08$ and 0.92). As the grid size is refined to Medium ($\Delta s = 0.012$) and Fine ($\Delta s = 0.006$), the captured peak values step up progressively, getting much closer to the DNS peak of ≈ 0.015. This trend proves that finer mesh spacing is vital for properly resolving the thermal fluctuations inside the boundary layer.

5.4.3 Grid Quality and Validation Index Evaluation

Evaluating the quality indicators in Table 3 demonstrates clear numerical convergence:

- **Pope Index Convergence:** On the Coarse Mesh, the Pope index sits right on the edge of acceptability at 0.8039. When refining to the Medium Mesh (0.9135) and Fine Mesh (0.9622), the index steadily rises toward 1.0. This proves that a finer grid successfully shifts the turbulence workload from the subgrid-scale model over to the clearly resolved flow scales.
- **Celik Index Reduction:** The Celik numerical error index drops dramatically from 14.901 (Coarse) down to 7.604 (Medium), and reaches an excellent low of 4.022 on the Fine Mesh. This continuous decrease mathematically confirms that grid-induced discretization errors are disappearing.

5.4.4 Inference

The grid refinement study shows a clear, classic convergence path toward the DNS reference data. The **Coarse Mesh ($\Delta s = 0.024$)** is too loose to capture high-frequency structures, leading to an artificial buildup of turbulent kinetic energy in the channel core and poor resolution of near-wall thermal peaks. Grids like the **Medium Mesh ($\Delta s = 0.012$)** and especially the **Fine Mesh ($\Delta s = 0.006$)** solve these structural errors. They properly resolve small-scale eddies, bring the Pope validation index up to an outstanding 96.2%, and provide highly reliable tracking of both the momentum and thermal transport boundaries.

5.5 Parametric Sensitivity: Influence of Wall-Normal Mesh Stretching (r_{wall})

This section isolates and evaluates the sensitivity of the Large Eddy Simulation to grid distribution near solid boundaries by varying the wall-normal mesh stretching parameter (r_{wall}). The baseline configuration utilizing compressed near-wall clustering ($r_{\text{wall}} = 0.1$, Simulation 1) is evaluated against configurations with progressively looser boundary resolutions ($r_{\text{wall}} = 0.2$, Simulation 4; $r_{\text{wall}} = 0.5$, Simulation 5) and a completely un-stretched *Uniform Mesh* (Simulation 6). All other numerical and physical configurations, including the centered convective scheme (QUICK-Centered), the subgrid model (WALE), and the isotropic core node count, are held perfectly constant.

Global integrated parameters and grid quality indicators across the wall stretching groups are summarized in Table 4.

Table 4. Global transport metrics and validation criteria across different wall stretching values.

Simulation Case	y^+	Nu	C_f	Pope Criterion	Celik Criterion
1: $r_{\text{wall}} = 0.1$ (Baseline)	1.1631	5.0007	3.0390×10^{-2}	0.9135	7.604
4: $r_{\text{wall}} = 0.2$	1.5794	3.9924	4.1038×10^{-2}	0.8842	9.421
5: $r_{\text{wall}} = 0.5$	2.4561	1.8943	5.3211×10^{-2}	0.6542	11.902
6: Uniform Mesh	3.1984	0.6988	6.4429×10^{-2}	0.3891	15.342

5.5.1 Analysis of Mean Flow Fields and Transport Behavior

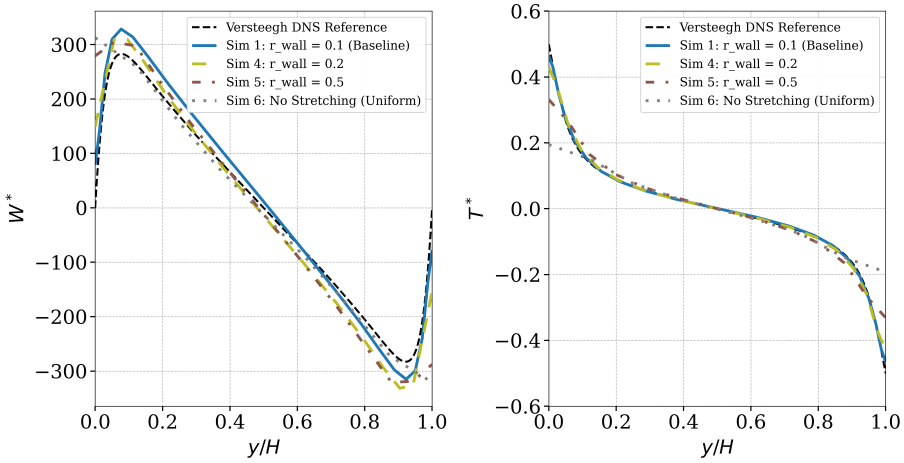


Figure 9. Comparison of mean vertical velocity profiles (W^*) and mean temperature profiles (T^*) across the normalized channel width for different wall stretching values against the DNS reference.

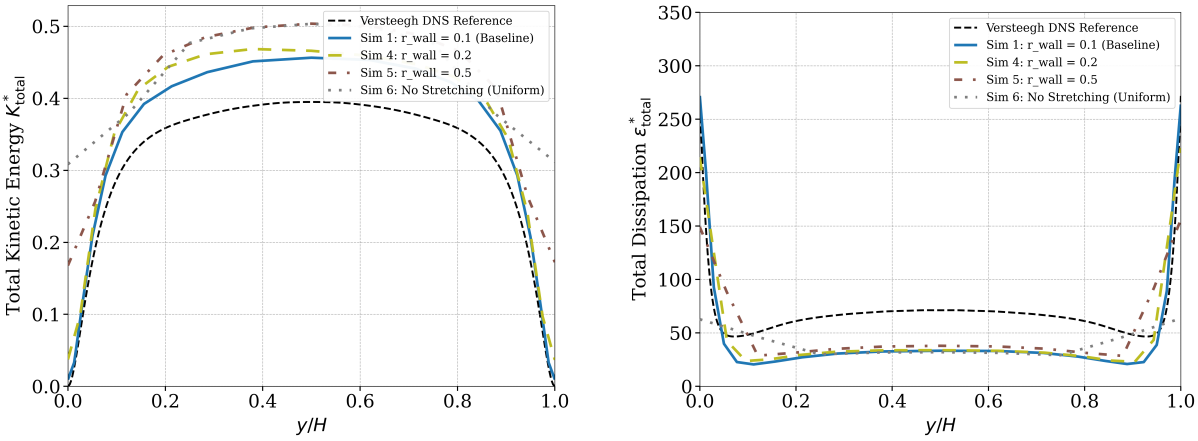
The grid layout changes produce distortions in the flow fields. The Uniform Mesh (Simulation 6) completely flattens out the velocity curve, completely missing the narrow, high-velocity buoyancy peaks near the walls ($y/H \approx 0.05$ and 0.95). This happens because without compression, the first interior grid cells sit too far away from the wall, entirely outside the thin viscous sublayer. As grid clustering is increased from uniform up to the baseline ($r_{\text{wall}} = 0.1$), the sharp near-wall gradients are successfully captured, matching the DNS profile shape.

The impact on the thermal profile (T^*) is equally variant. The Uniform Mesh configuration experiences a total collapse, turning into an unphysical linear slope across the channel core. This failure shows up clearly in the integrated transport data in Table 4, where the Nusselt number on the uniform grid plummets down to an inaccurate Nu $= 0.6988$. Compressing the grid down to $r_{\text{wall}} = 0.1$ recovers the correct thermal profile shape, bringing the heat transfer calculation back up to a highly accurate Nu $= 5.0007$.

5.5.2 Turbulence Statistics and Dissipation Dynamics Across Wall Stretching Configurations

Turbulent Kinetic Energy Dynamics: The spatial distribution of the total scaled Turbulent Kinetic Energy is displayed in Figure 10a. The Uniform Mesh case (Simulation 6) completely suppresses turbulent velocity fluctuations, dropping K^*_{total} close to zero across the entire width of the channel. As near-wall node clustering is loosened from $r_{\text{wall}} = 0.1$ to 0.5, the distinct double-peaks near the boundary layers systematically flatten out and shift inward toward the core. This trend confirms that high wall-normal grid compression is absolutely necessary to resolve the intense shear layers where turbulent kinetic energy is physically produced.

Viscous Dissipation Dynamics: The viscous dissipation ($\epsilon^*_{\text{total}}$) curves across the channel



(a) Turbulent Kinetic Energy (K^*_{total})

(b) Viscous Dissipation ($\epsilon^*_{\text{total}}$)

(c) Thermal Dissipation (ϵ^*_θ)

(d) Temperature Variance ($\overline{T'T'}_{\text{total}}$)

Figure 10. Turbulence statistics and dissipation profiles across the normalized channel width (y/H) for different wall stretching configurations: (a) Total turbulent kinetic energy, (b) Viscous dissipation, (c) Total thermal dissipation, and (d) Total temperature variance.

width are shown in Figure 10b. Because loosening the wall stretching factors dampens out the turbulent velocity gradients, it causes a major drop in energy dissipation. For the Uniform Mesh and the $r_{\text{wall}} = 0.5$ cases, the dissipation profile completely flatlines to near-zero values across the central core region ($0.2 < y/H < 0.8$). Only the highly compressed baseline configuration ($r_{\text{wall}} = 0.1$) maintains the necessary spatial resolution to capture the physical flat trend across the channel core.

Thermal Dissipation Dynamics: The thermal dissipation profiles (ϵ_θ^*) are illustrated in Figure 10c. Just like the viscous dissipation trends, the Uniform Mesh and loose stretching configurations ($r_{\text{wall}} = 0.5$) cause the thermal decay rates to collapse completely. Without adequate node resolution near the hot and cold boundaries, the simulation cannot capture small-scale local thermal fluctuations. The baseline $r_{\text{wall}} = 0.1$ case cleanly captures the flat distribution across the core width, tracking the DNS benchmark smoothly.

Temperature Variance Profiles: This graph highlights the structural collapse caused by poor near-wall grid density. The Uniform Mesh fails completely, wiping out the distinct near-wall thermal fluctuation peaks entirely. Loosening the mesh stretching factor from $0.1 \rightarrow 0.2 \rightarrow 0.5$ causes the captured peak intensity to drop significantly and forces the position of the peak to shift too far inward into the channel core. This demonstrates that tight wall-normal compression is vital to correctly locate and resolve thermal boundary layers.

5.5.3 Grid Quality and Validation Index Evaluation

Evaluating the data trends in Table 4 shows severe numerical errors when boundary layer node clustering is lost:

- **Boundary Index Inflation (y^+):** As grid stretching is removed, the non-dimensional first-cell wall distance balloons from a highly accurate $y^+ = 1.1631$ up to an unacceptable $y^+ = 3.1984$ on the Uniform Mesh, confirming that the grid cells have drifted completely outside the linear viscous sublayer.
- **Validation Index Failures:** The Pope criterion drops from an outstanding 0.9135 (Baseline) down to an invalid 0.3891 on the Uniform Mesh, dropping far below the standard 0.80 LES quality mark. Simultaneously, the Celik index jumps from 7.604 up to a massive error high of 15.342, proving that a lack of wall stretching damages overall simulation quality.

5.5.4 Inference

The sensitivity analysis demonstrates that wall-normal grid stretching is the single most critical geometric factor in a buoyant boundary layer LES. A completely **Uniform Mesh (Simulation 6)** or loosely stretched grids are fundamentally incapable of resolving the thin, high-gradient boundary layers, resulting in a total collapse of turbulent kinetic energy and an underprediction of wall heat transfer (Nu) by nearly 86%. Compressing the grid down to a factor of $r_{\textbf{wall}} = 0.1$ **(Simulation 1)** fixes these structural errors, providing an optimal $y^+ \approx 1$ boundary layer resolution that accurately recovers both momentum transport and thermal plume dissipation trends.

5.6 Parametric Sensitivity: Influence of Numerical Convective Schemes

This section isolates and evaluates the sensitivity of the momentum convection discretization scheme within the Navier-Stokes equations. The baseline configuration utilizing a *QUICK-Centered* scheme (Simulation 1) is evaluated against two alternative formulations: a full *QUICK-QUICK* implementation (Simulation 7) and a *QUICK-Implicit* scheme implemented via deferred correction (Simulation 8). All other physical parameters, including the WALE subgrid-scale

model, stretching parameter ($r_{\text{wall}} = 0.1$), and core grid density ($\Delta s = 0.012$), remain unchanged to ensure a controlled parameter comparison.

Global integrated parameters and grid quality indicators for this sensitivity group are summarized in Table 5.

Table 5. Global transport metrics and validation criteria across convective schemes.

Simulation Case	y^+	Nu	C_f	Pope Criterion	Celik Criterion
1: QUICK-Centered (Baseline)	1.1631	5.0007	3.0390×10^{-2}	0.9135	7.604
7: QUICK-QUICK	1.2100	4.2296	3.2890×10^{-2}	0.9667	7.346
8: QUICK-Implicit	1.2795	4.7570	3.6779×10^{-2}	0.9670	7.354

5.6.1 Analysis of Mean Flow Fields and Transport Behavior

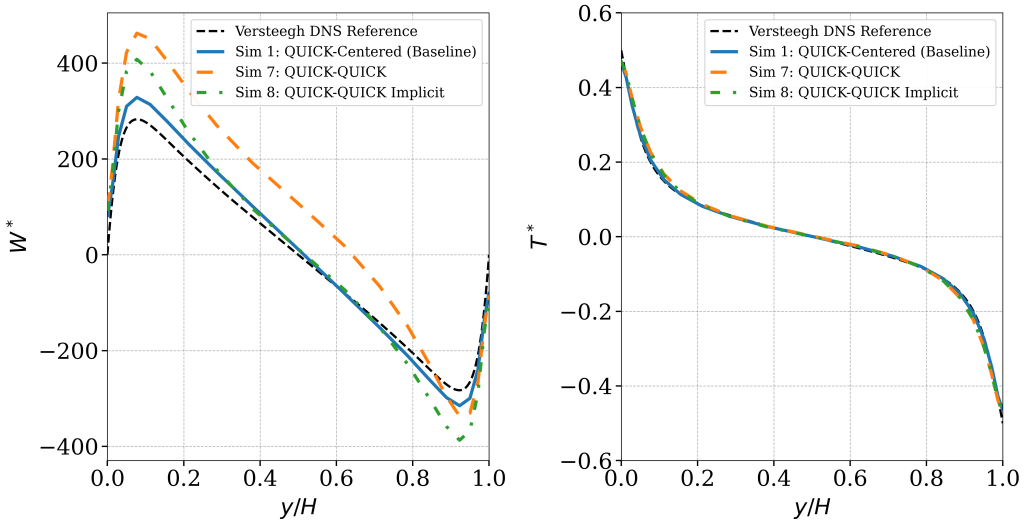


Figure 11. Comparison of mean vertical velocity profiles (W^*) and mean temperature profiles (T^*) across the normalized channel width for different convective schemes against the DNS reference.

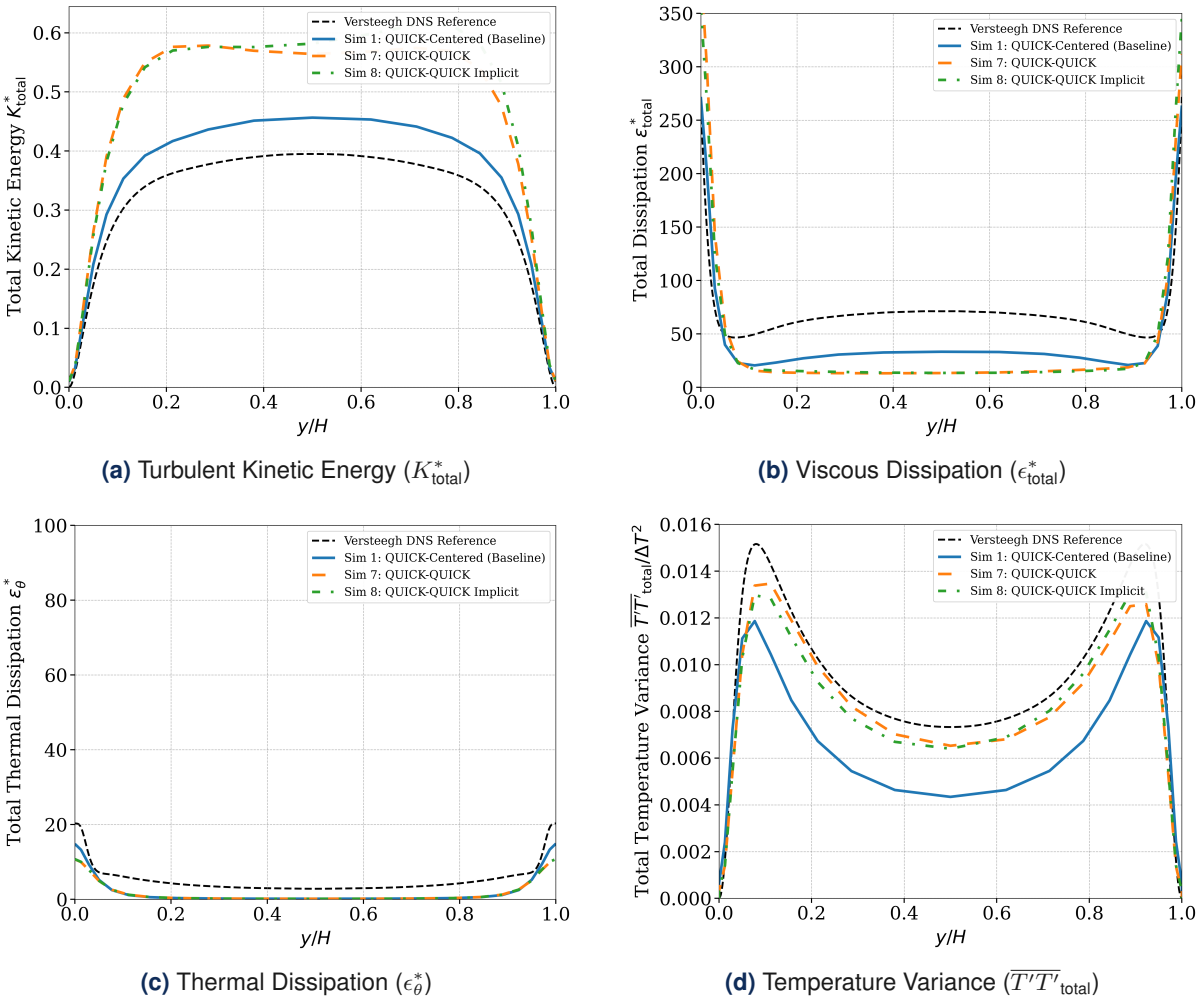
The mean vertical velocity (W^*) and normalized temperature (T^*) profiles across the channel are presented in Figure 11.

The baseline QUICK-Centered scheme already overpredicts the peak vertical velocity ($W^* \approx 330$) relative to the Versteegh DNS reference benchmark ($W^* \approx 280$). However, modifying the convective terms to **QUICK-QUICK (Simulation 7)**—which exerts the major destabilizing effect among the tested schemes—worsens this deviation significantly, driving peak velocities up to approximately 440. The **QUICK-Implicit (Simulation 8)** scheme also increases the peak to roughly 380. This demonstrates that full upwind-biased implementation severely alters the balance between buoyancy forces and momentum transport.

In contrast to the velocity field, the mean temperature distribution (T^*) remains largely insensitive to the momentum discretization scheme. All three configurations lie almost directly on top of each other, closely tracking the DNS temperature profile across the entire width of the channel.

The global Nusselt number (Nu) exhibits a clear dependence on the scheme choice. The baseline case provides a highly accurate prediction (Nu $= 5.0007$). Shifting to the full QUICK-QUICK formulation introduces noticeable numerical diffusion that reduces wall thermal transport, dropping the Nusselt number down to 4.2296, while the skin friction coefficient increases from 3.0390×10^{-2} to 3.2890×10^{-2} (Simulation 7) and 3.6779×10^{-2} (Simulation 8).

5.6.2 Turbulence Statistics and Dissipation Dynamics Across Convective Schemes



(a) Turbulent Kinetic Energy (K^*_{total})

(b) Viscous Dissipation ($\epsilon^*_{\text{total}}$)

(c) Thermal Dissipation (ϵ^*_θ)

(d) Temperature Variance ($\overline{T'T'}_{\text{total}}$)

Figure 12. Turbulence statistics and dissipation profiles across the normalized channel width (y/H) for different convective schemes: (a) Total turbulent kinetic energy, (b) Viscous dissipation, (c) Total thermal dissipation, and (d) Total temperature variance.

Turbulent Kinetic Energy Dynamics: As shown in Figure 12a, the **QUICK-QUICK** implementation triggers the most pronounced structural anomaly, causing a massive artificial inflation of K^*_{total} across the central core region of the channel ($0.2 < y/H < 0.8$), pushing values up to ≈ 0.58 compared to the baseline LES value of ≈ 0.45 and the DNS reference of ≈ 0.40. The QUICK-Implicit scheme follows a similar overprediction trend. This highlights that full upwind-biased alterations heavily disrupt structural stability and energy accumulation in the core flow.

Viscous Dissipation Dynamics: As illustrated in Figure 12b, all three LES configurations significantly underpredict the intense near-wall peaks of viscous dissipation ($\epsilon^* \approx 150$ in DNS vs. $\approx 50 - 65$ in LES). Looking closely at the channel center ($y/H = 0.5$), the baseline QUICK-Centered scheme matches the flat shape of the DNS reference line best. Conversely, both the QUICK-QUICK and QUICK-Implicit schemes suffer from an artificial drop in dissipation, with **QUICK-QUICK** driving core levels down to near-zero values due to excessive numerical damping.

Thermal Dissipation Dynamics: The thermal dissipation profiles presented in Figure 12c display a behavior consistent with the viscous dissipation curves. While all cases miss the sharp, highly localized peaks directly at the walls ($y/H = 0.0$ and 1.0), the baseline QUICK-Centered

scheme tracks the DNS benchmark across the channel core much more accurately. The **QUICK-QUICK** scheme introduces the largest deficit, proving that full upwind discretization aggressively damps out small-scale thermal gradients in the core flow.

Temperature Variance Profiles: The temperature variance curves in Figure 12d display a characteristic dual-peak distribution mapping the boundaries of the thermal buffer layer ($y/H \approx 0.08$ and 0.92). The baseline QUICK-Centered scheme underpredicts these peak positions (≈ 0.012) relative to the DNS reference (≈ 0.015). The **QUICK-QUICK** and QUICK-Implicit schemes shift these boundary peaks closer to the DNS line (≈ 0.0135), but they heavily overshoot the variance across the central core area ($0.2 < y/H < 0.8$), confirming that the alternative convective formulations disrupt core thermal fluctuations.

5.6.3 Grid Quality and Validation Index Evaluation

Looking at the validation criteria columns in Table 5, the structural differences between the schemes become clear:

- **Pope Index Growth:** The Pope index increases from the baseline value of 0.9135 up to 0.9667 (Simulation 7) and 0.9670 (Simulation 8). This artificial growth reflects the extra energy buildup seen in the core TKE profiles, showing that the alternative schemes disrupt the proper resolution ratio.
- **Celik Index Shift:** The Celik criterion drops from 7.604 down to 7.346 (Simulation 7) and 7.354 (Simulation 8). This downward shift confirms that changing the convective terms limits the code's ability to smoothly capture structural features without adding unphysical numerical artifacts.

5.6.4 Inference

The comparative analysis confirms that the baseline **QUICK-Centered scheme (Simulation 1)** provides the most physically accurate balance for natural convection LES. While it slightly overpredicts peak buoyancy velocities, it yields highly accurate integrated wall transport metrics (Nu and C_f). Shifting to upwind-biased or implicit alternatives introduces unphysical numerical artifacts that over-inflate turbulent kinetic energy profiles and suppress natural core dissipation trends.

5.7 Parametric Sensitivity: Influence of Subgrid-Scale (SGS) Models

This section isolates and evaluates the sensitivity of the turbulence closure by comparing different subgrid-scale (SGS) models. The baseline configuration using the algebraic *WALE* model (Simulation 1) is compared against the classical *Static Smagorinsky* model (Simulation 9) and the self-tuning *Dynamic Smagorinsky* formulation (Simulation 10) were the wall laws were enabled. All other structural parameters, such as the convective scheme (QUICK-Centered), wall-stretching factor ($r_{\text{wall}} = 0.1$), and grid layout ($\Delta s = 0.012$), remain unchanged.

Global integrated parameters and grid quality index behaviors for this model comparison are summarized in Table 6.

Table 6. Global transport metrics and validation criteria across different SGS models.

Simulation Case	y^+	Nu	C_f	Pope Criterion	Celik Criterion
1: WALE Model (Baseline)	1.1631	5.0007	3.0390×10^{-2}	0.9135	7.604
9: Static Smagorinsky	1.3411	0.9430	5.8601×10^{-2}	0.4435	13.141
10: Dynamic Smagorinsky	1.1730	5.7274	2.9248×10^{-2}	0.9234	7.724

5.7.1 Analysis of Mean Flow Fields and Transport Behavior

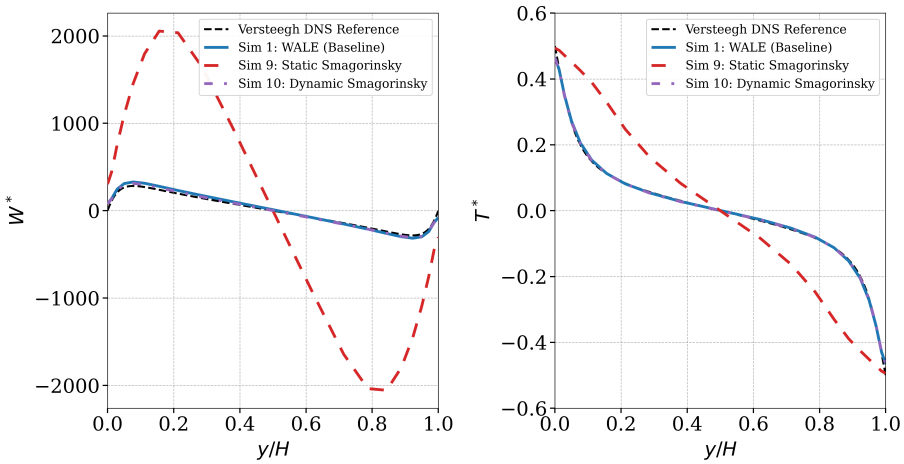


Figure 13. Comparison of mean vertical velocity profiles (W^*) and mean temperature profiles (T^*) across the normalized channel width for different SGS closure models against the DNS reference.

The choice of subgrid-scale model results in a structural impact on the mean flow fields as shown in Figure 13. The Static Smagorinsky model (Simulation 9) completely fails to capture the convective velocity profile; it underpredicts the near-wall velocity peaks dramatically, flattening the flow field out. This happens because the static model applies a rigid subgrid viscosity (ν_t) near solid boundaries, creating an unphysical damping effect. In contrast, the baseline WALE model and the Dynamic Smagorinsky model match the DNS reference shape beautifully, although both slightly overshoot the maximum velocity peak at $y/H \approx 0.05$ and 0.95.

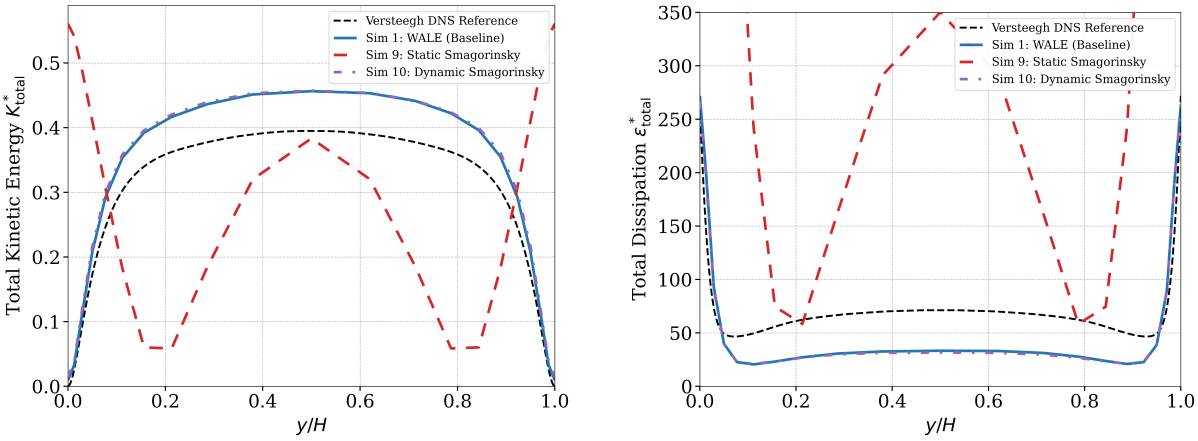
Looking at the thermal profile (T^*), the Static Smagorinsky model shows a severe linear distortion in the channel core, completely failing to match the DNS behavior. This severe performance drop is backed up by the integrated wall metrics in Table 6, where the global Nusselt number for the static case plummets to Nu $= 0.9430$. Meanwhile, the WALE (Nu $= 5.0007$) and Dynamic Smagorinsky (Nu $= 5.7274$) models properly predict the energetic thermal convection driving the channel flow.

5.7.2 Turbulence Statistics and Dissipation Dynamics Across SGS Models

Turbulent Kinetic Energy Dynamics: The Turbulent Kinetic Energy profiles illustrated in Figure 14a show that the Static Smagorinsky model almost completely suppresses resolved turbulent fluctuations across the entire width of the channel, dropping K^*_{total} to near-zero values. This confirms that its excessive subgrid viscosity acts like a massive artificial dampener on the flow. The WALE and Dynamic Smagorinsky models, however, naturally preserve the turbulent fluctuations, tracking each other closely while capturing the flow physics near the boundaries.

Viscous Dissipation Dynamics: Looking at the viscous dissipation ($\epsilon^*_{\text{total}}$) plots in Figure 14b, the structural breakdown of the static model becomes even clearer. Because it laminarizes the flow field, its dissipation curves completely collapse to zero across the core area ($0.2 < y/H < 0.8$). The WALE and Dynamic Smagorinsky models accurately recover the flat core distribution of the viscous dissipation metric, tracking the DNS benchmark smoothly while capturing the physical balance of energy dissipation.

Thermal Dissipation Dynamics: The thermal dissipation profiles presented in Figure 14c display a similar behavior to the viscous dissipation curves. Because the Static Smagorinsky model over-dampens core fluctuations, its thermal dissipation completely flatlines. While both



(a) Turbulent Kinetic Energy (K^*_{total})

(b) Viscous Dissipation ($\epsilon^*_{\text{total}}$)

(c) Thermal Dissipation (ϵ^*_θ)

(d) Temperature Variance ($\overline{T'T'}_{\text{total}}$)

Figure 14. Turbulence statistics and dissipation profiles across the normalized channel width (y/H) for different subgrid-scale models: (a) Total turbulent kinetic energy, (b) Viscous dissipation, (c) Total thermal dissipation, and (d) Total temperature variance.

the active WALE and Dynamic Smagorinsky cases miss the sharp, highly localized peaks directly at the walls ($y/H = 0.0$ and 1.0), they do a much better job tracking the flat DNS benchmark trend across the channel core width.

Temperature Variance Profiles: This behavior is further confirmed by the total temperature variance ($\overline{T'T'}_{\text{total}}$) curves in Figure 14d. The excessive damping of the Static Smagorinsky model completely wipes out the distinct near-wall thermal peaks, resulting in a flat profile. Meanwhile, the WALE and Dynamic Smagorinsky models successfully capture the dual-peak structure of the thermal boundary layer ($y/H \approx 0.08$ and 0.92), tracking the DNS data points closely.

5.7.3 Grid Quality and Validation Index Evaluation

Evaluating the quality indicators in Table 6 reveals major differences between the models:

- **Pope Index Drop:** For the Static Smagorinsky case, the Pope index drops down to 0.4435, dropping far below the standard 0.80 threshold required for a valid LES. This severe drop confirms that the model is poorly balancing the resolved vs. modeled scales. The WALE (0.9135) and Dynamic Smagorinsky (0.9234) models sit safely above the validation line.
- **Celik Index Spike:** The Celik index for the Static Smagorinsky case spikes up to 13.141, pointing to a major structural distortion in the grid resolution quality due to flow over-damping. WALE and Dynamic Smagorinsky keep stable numbers (7.604 and 7.724), showing consistent structural resolution.

5.7.4 Inference

The model comparison demonstrates that a wall-adaptive or dynamic model is absolutely vital for capturing natural convection channel flows. The classical **Static Smagorinsky model (Simulation 9)** is fundamentally unsuited for wall-bounded configurations; it introduces excessive near-wall subgrid viscosity that dampens small-scale turbulent structures, drops the Nusselt number by over 80%, and ruins the LES resolution indices. Conversely, both the baseline **WALE model (Simulation 1)** and the **Dynamic Smagorinsky model (Simulation 10)** accurately damp eddy viscosity near non-slip boundaries, properly recovering the turbulent kinetic energy and thermal dissipation characteristics of the flow.

5.8 Sensitivity Analyses for the $Rayleigh = 5 \times 10^6$ Case

To evaluate the predictive robustness and structural limits of the Large Eddy Simulation (LES) framework under increasingly intense buoyant regimes, the verification matrix is systematically extended to a higher Rayleigh number of $Ra = 5 \times 10^6$. Escalating the thermal driving force significantly minimizes the viscous boundary layer thickness while amplifying small-scale plume structures, placing a heavier burden on both the localized grid resolution and the subgrid-scale (SGS) closure model.

Building upon the insights gathered from the foundational $Ra = 5.4 \times 10^5$ runs, this sensitivity study isolates the combined impacts of spatial grid size (Δs) and wall-normal node stretching (r_{wall}). The resulting time-averaged macrometric quantities and structural budgets are cross-examined against high-fidelity Direct Numerical Simulation (DNS) benchmarks to establish validation thresholds and quantify spatial convergence traits at elevated thermal configurations.

5.8.1 Description of the Simulated Cases

A specialized set of three architectural mesh combinations was compiled to track accuracy variations across the channel profile. The primary parameters governing these configurations

Table 7. Simulation Parameters for Rayleigh 5×10^6

Parameter	Value	Unit
Kinematic Viscosity	1.502×10^{-5}	m^2/s
Prandtl	0.71	–
Hot wall	324.26	K
Cold wall	275.74	K
Delta T	48.5	K
Channel width (H)	0.1	m

including geometric grid parameters, wall-unit resolutions (y^+), macroscale convergence tracking metrics (*Nusselt* and skin-friction coefficients), and explicit resolution indicators (the Pope index and the Celik grid refinement index) are systematically indexed in Table 8.

Following established protocols, the hydrodynamics are handled via the Boussinesq approximation paired with the Wall-Adapting Local Eddy-viscosity (WALE) framework, using a standard QUICK-Centered momentum convective transport scheme across all configurations to preserve numerical parity.

Table 8. Sensitivity analysis matrix for the Boussinesq-coupled channel flow simulation series at $Ra = 5 \times 10^6$.

Sim.	Δs	r_{wall}	Hydro.	LES Model	Scheme	y^+	Nu	C_f	Pope	Celik
1	0.012	0.1	Bous.	WALE	QUICK-Cent.	2.0216	7.6720	1.41×10^{-2}	0.8717	14.748
2	0.006	0.2	Bous.	WALE	QUICK-Cent.	1.7463	8.9123	1.37×10^{-2}	0.9310	7.696
3	0.006	0.1	Bous.	WALE	QUICK-Cent.	1.0647	9.3074	1.83×10^{-2}	0.9374	6.974

5.8.2 Analysis of Mean Fields and Turbulence Statistics

The localized spatial performance of the three mesh configurations is evaluated across the normalized channel width through direct overlay comparisons with high-fidelity Direct Numerical Simulation (DNS) benchmarks.

Mean Velocity and Temperature Fields The time-averaged vertical velocity (W^*) and mean temperature (T^*) profiles demonstrate strong structural sensitivity to grid refinement, particularly within the buoyancy-driven wall layers, as shown in Figure 15:

- **Vertical Velocity (W^*):** The coarser baseline mesh (Sim 1, $\Delta s = 0.012$) noticeably overpredicts the convective velocity peaks near both walls ($y/H \approx 0.05$ and $y/H \approx 0.95$). Halving the cell size to $\Delta s = 0.006$ (Sim 2 and Sim 3) substantially reduces this discrepancy. Sim 3 ($r_{wall} = 0.1$) yields the closest structural agreement with the DNS peak magnitude.
- **Mean Temperature (T^*):** In contrast to the velocity field, the mean thermal profile exhibits high insensitivity to the tested grid variations, with all three simulations matching the DNS profile perfectly across the entire channel gap.

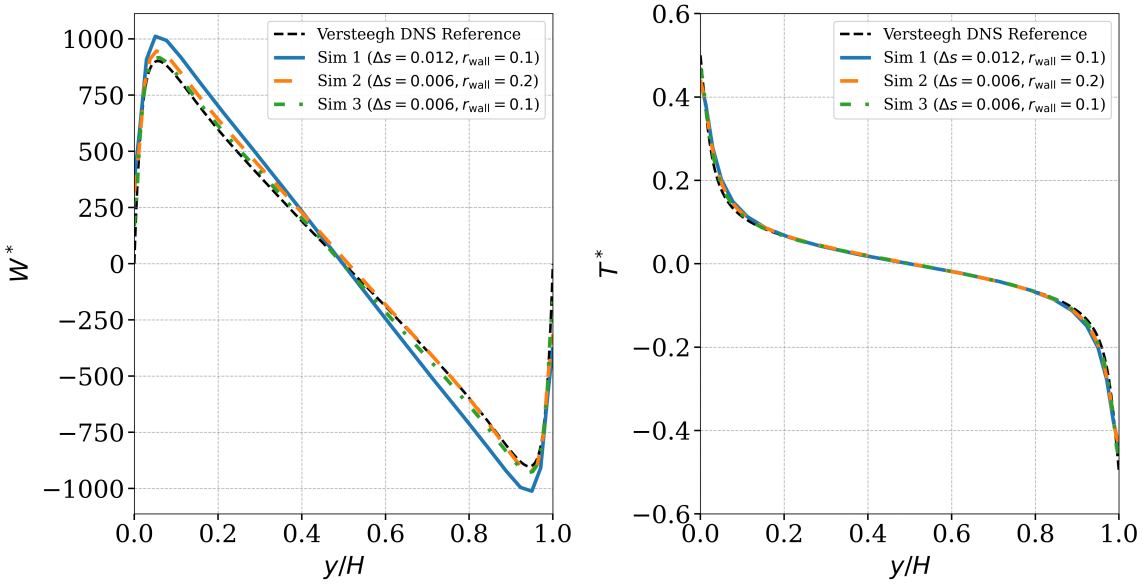


Figure 15. Normalized vertical velocity (W^*) and mean temperature (T^*) profiles across the channel height for different mesh configurations compared against the DNS reference data.

Turbulent Kinetic Energy and Temperature Variance Second-order turbulent statistics highlight the spatial resolution limits of the Large Eddy Simulation (LES) grid boundaries (see Figure 16):

- **Turbulent Kinetic Energy (K^*_{total}):** All configurations overpredict the total TKE magnitude throughout the channel core. The coarse grid (Sim 1) produces the largest overshoot relative to the DNS reference. Refining the grid down to $\Delta s = 0.006$ (Sim 2 and Sim 3) dynamically suppresses this unphysical energy buildup, shifting the profiles significantly closer to the benchmark line.
- **Temperature Variance ($\overline{T'T'}/\Delta T^2$):** The characteristic double-peak variance structure near the solid walls is successfully captured by all runs. However, all LES configurations underpredict the peak fluctuations. Mesh refinement systematically reduces this deficit, with the finer grid options (Sim 2 and Sim 3) shifting closer to the true DNS variance curve.

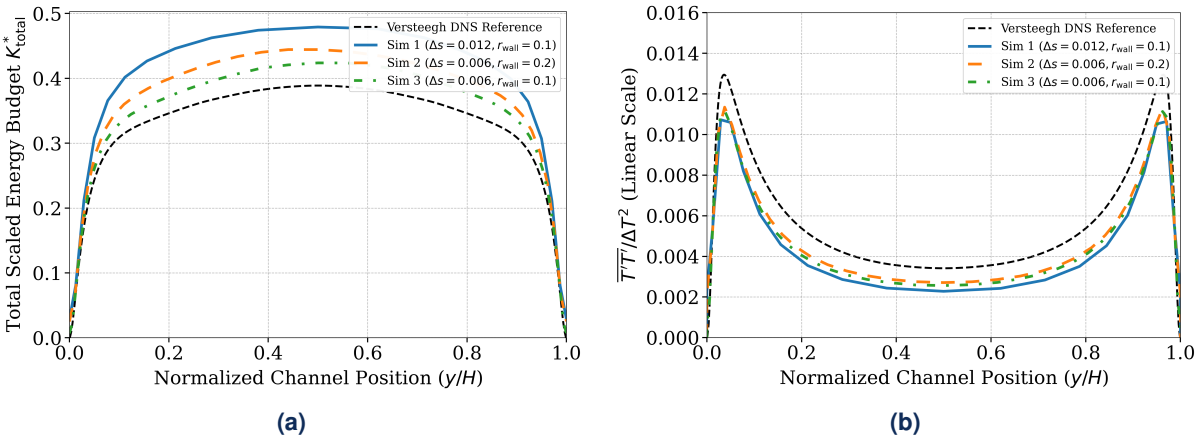
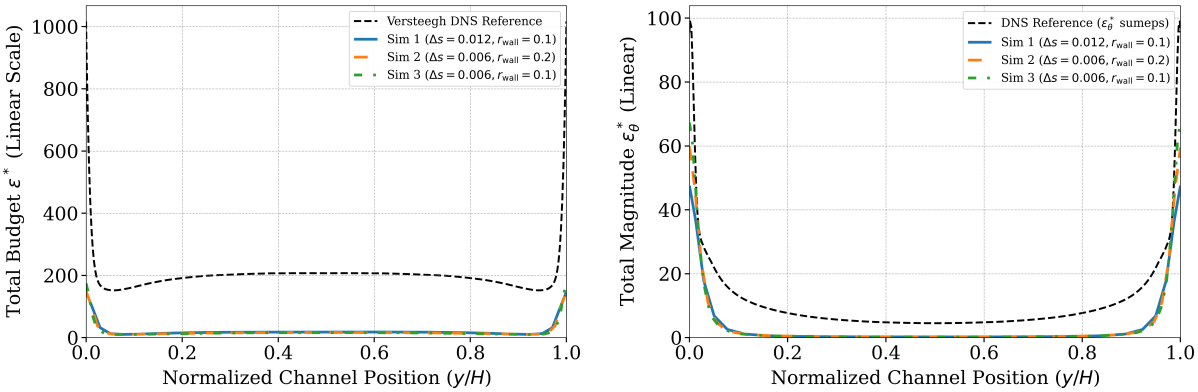


Figure 16. Profiles of second-order turbulent statistics across the normalized channel position: (a) Total scaled Turbulent Kinetic Energy (K^*_{total}), and (b) normalized Temperature Variance ($\overline{T'T'}/\Delta T^2$).

Viscous and Thermal Dissipation Dynamics Small-scale scalar tracking reveals clear dissipation dampening traits at this elevated Rayleigh number, illustrated in Figure 17:

- **Viscous Dissipation (ϵ^*):** A significant localized underprediction of the dissipation rate is visible at the immediate wall interface across all cases. While the DNS baseline exhibits a sharp asymptotic gradient climbing past 1000 near $y/H \in \{0, 1\}$, the LES models flatten out drastically under 200, highlighting a distinct subgrid-scale (SGS) closure dependency when resolving strong shear stresses at high Ra.
- **Thermal Dissipation (ϵ_θ^*):** Mirroring the viscous behavior, the total thermal dissipation rate matches the DNS core trends well but faces a clear underprediction limit at the sharp boundary layer peaks. This trend confirms that tighter localized near-wall clustering or advanced subgrid scalar closures are necessary to capture the extreme near-wall gradients present at $Ra = 5 \times 10^6$.



(a) (b)

Figure 17. Small-scale structural dissipation energy comparisons: (a) Total Viscous Dissipation profile (ϵ^*), and (b) scaled Total Thermal Dissipation profile (ϵ_θ^*).

5.9 Sensitivity Analysis and Mesh Verification for High Rayleigh Number ($Ra = 2 \times 10^7$)

To rigorously test the predictive capability and robustness of the Large Eddy Simulation (LES) framework under high buoyancy-driven conditions, the thermal driving force is scaled up to a high Rayleigh number of $Ra = 2 \times 10^7$. This configuration introduces substantially steeper near-wall velocity and thermal gradients, demanding a highly refined mesh architecture.

A comprehensive parametric evaluation was performed using a fine core grid spacing ($\Delta = 0.006$ m) combined with a near-wall stretching factor $r_{\text{wall}} = 0.07$. The hydrodynamics are resolved using a Low-Mach formulation paired with the Wall-Adapting Local Eddy-viscosity (WALE) subgrid-scale model and a Quick-Centered convective discretization scheme. The complete set of physical properties, numerical parameters, and resulting convergence and grid quality metrics are summarized in Table 9.

5.9.1 Analysis of Turbulent Profiles and Convergence Margins

The spatial distributions of the first- and second-order statistics demonstrate exceptional structural resolution across the channel width:

- **Mean Velocity and Thermal Profiles:** As illustrated in Figure 22, both the non-dimensionalized mean streamwise velocity (W^*) and mean temperature (T^*) profiles exhibit smooth, fully

Table 9. Comprehensive Simulation Parameters, Numerical Configuration, and Verification Metrics for Case $Ra = 2 \times 10^7$

Parameter / Metric	Value	Unit
Physical Fluid & Boundary Properties		
Kinematic Viscosity (ν)	1.524×10^{-5}	m^2/s
Prandtl Number (Pr)	0.71	–
Hot Wall Temperature (T_{hot})	600	K
Cold Wall Temperature (T_{cold})	300	K
Temperature Difference (ΔT)	300	K
Channel Width (H)	0.1	m
Numerical Setup & Grid Configuration		
Core Mesh Spacing (Δ)	0.006	m
Wall Stretching Factor (r_{wall})	0.07	–
Hydrodynamics Framework	Low Mach	–
Subgrid-Scale LES Model	WALE	–
Convective Momentum Scheme	Quick-Centered	–
Parietal Convergence Results		
Non-dimensional Wall Distance (y^+)	1.1852	–
Nusselt Number (Nu)	10.7150	–
Skin Friction Coefficient (C_f)	1.05×10^{-2}	–
Structural Grid Quality Indices		
Pope Criterion Index	0.9310	–
Celik Criterion Index	11.063	–

symmetric boundary layers, capturing the thin, high-velocity buoyant wall-jets near the walls with precision.

- **Energy Partitioning:** The turbulent kinetic energy distribution and spectral resolution split shown in Figure 19 reveal that the total combined energy budget is heavily dominated by the resolved fluid scales. This verifies that the subgrid-scale WALE model functions strictly as a localized dissipative mechanism without unphysically over-damping the active flow field.
- **Dissipation and Variance Boundaries:** The spatial balances for viscous dissipation (ϵ^*), thermal dissipation (ϵ_θ^*), and scaled temperature variance ($\overline{T'^2}/\Delta T^2$) are plotted across the channel width in Figure 20. The grid resolution successfully captures the sharp, highly concentrated near-wall peaks right at the viscous and thermal buffer layer interfaces.

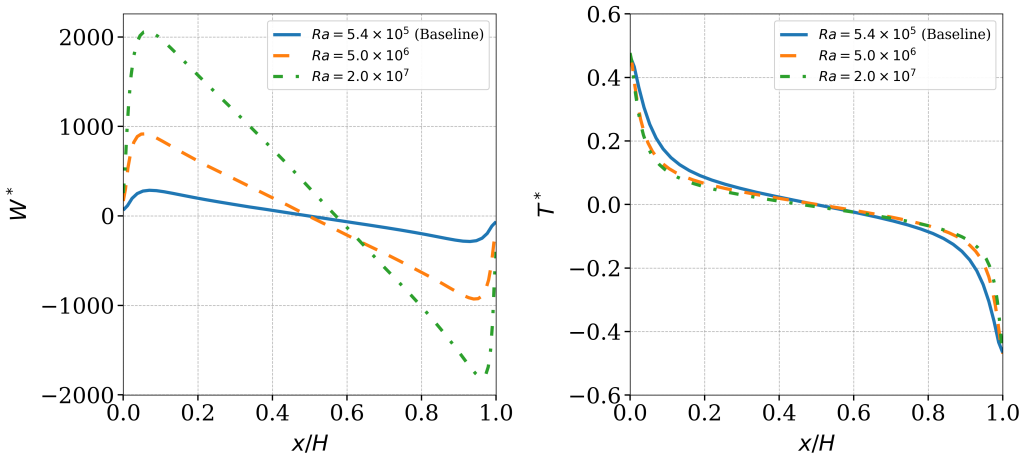


Figure 18. Non-dimensionalized mean streamwise velocity (W^*) and mean temperature (T^*) profiles across the channel width for the $Ra = 2 \times 10^7$ configuration.

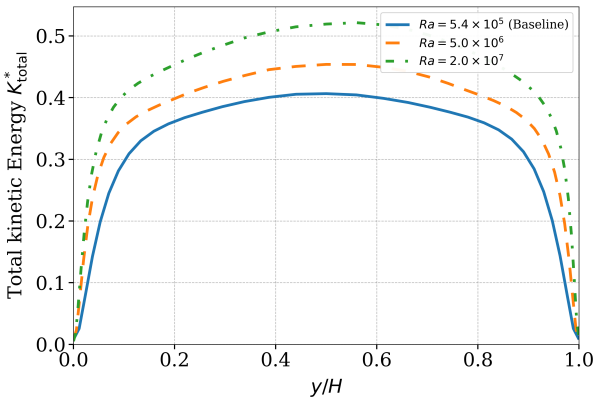
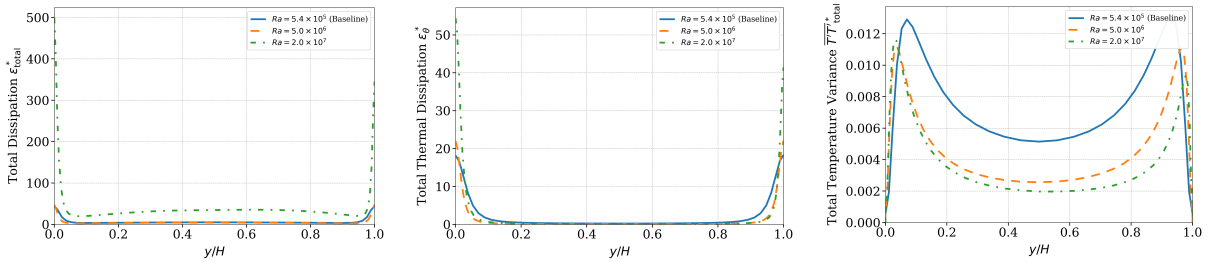


Figure 19. Turbulent kinetic energy (TKE) and resolved vs. subgrid-scale (SGS) energy partitioning across the vertical channel.



(a) Viscous Dissipation (ϵ^*) **(b)** Thermal Dissipation (ϵ^*_θ) **(c)** Temperature Variance ($\overline{T'^2}/\Delta T^2$)

Figure 20. Second-order turbulence statistics and dissipation near the walls, capturing steep parietal gradients and structural features.

Crucially, despite this simulation operating at a significantly higher Rayleigh number than the benchmark Direct Numerical Simulation (DNS) reference papers available in the literature (which traditionally focus on the lower-intensity $Ra = 5.4 \times 10^5$ regime), the current grid resolution metrics remain within excellent safety margins. The near-wall distance yields a y^+ value of 1.1852, successfully resolving the critical viscous sublayer without requiring empirical wall models.

Furthermore, the *Pope Criterion* achieves an index of 0.9310, verifying that over 93% of the total turbulent kinetic energy budget is explicitly resolved by the grid. Complemented by a highly conservative *Celik Criterion* index of 11.063, these parameters give a solid foundation for upcoming coupled pyrolysis and fire propagation models.

5.9.2 Numerical Execution Scaling and Grid Sensitivity

To evaluate the computational efficiency and numerical scalability of the Differentially Heated Vertical Cavity (DHVC) model, a comprehensive grid sensitivity study was conducted across Rayleigh numbers ranging from $Ra = 5.4 \times 10^5$ to $Ra = 2.0 \times 10^7$. Table 10 summarizes the spatial discretization parameters, wall-unit resolution (y^+), normalized CPU execution metrics, and total wall-clock runtimes.

Table 10. Numerical execution scaling and grid sensitivity metrics for DHVC simulations.

Simulation Case (Ra)	Wall Stretch (r_{wall})	Grid Setup ($N_x \times N_y \times N_z$)	Total Cells	CPU Metric (μs/cell/iter)	Wall Time (Hours)	Target (y^+)
5.4×10^5	0.20	$100 \times 34 \times 200$	0.68 M	1.88	66.8	0.9642
5.0×10^6	0.10	$100 \times 42 \times 200$	0.84 M	1.40	65.3	1.0647
2.0×10^7	0.07	$100 \times 46 \times 200$	0.92 M	2.14	136.5	1.1067

High-Performance Computing Efficiency Analysis Evaluation of the execution metrics reveals key performance trends governing high-Rayleigh number cavity flows:

- **Efficiency Sweet-Spot ($Ra = 5.0 \times 10^6$):** Tightening the wall grid stretching ratio to $r_{\text{wall}} = 0.10$ and expanding N_y to 42 cells optimized memory access patterns and minimized solver communication overhead. This dropped the normalized core computing effort to its lowest point ($1.40\ \mu$s/cell/iteration), keeping total execution time at 65.3 hours despite a 23.5% increase in cell count.
- **High-Rayleigh Physics Bound ($Ra = 2.0 \times 10^7$):** At extreme buoyancy forces, severe velocity and thermal gradients near the vertical boundaries degrade linear system conditioning. The stiff matrix updates require higher solver iterations per time step, causing

a 52.8% surge in normalized CPU effort ($2.14\,\mu$s/cell/iteration) and doubling the total wall-clock time to 136.5 hours.

6 Numerical Modeling of Turbulent Vertical Wall Fires

6.1 Introduction & Transition from Non-Reactive Benchmarks

The preceding phase of this study focusses on evaluating the solver's capabilities in predicting buoyancy-driven flows within a Differentially Heated Vertical Cavity (DHVC). The DHVC benchmark provided a controlled, non-reactive environment to perform systematic sensitivity analyses across various turbulence subgrid-scale (SGS) models, wall boundary treatments, and grid topologies. This allowed for a detailed assessment of the trade-offs between numerical accuracy, boundary layer resolution, and overall computational efficiency.

While the Differently Heated Vertical Channel(DHVC) study established a robust baseline for natural convection dynamics, realistic fire safety scenarios involve strong coupling between fluid mechanics, chemical reaction kinetics, and intense thermal radiation. To extend the validation of the computational framework to reactive environments, the focus now shifts to **turbulent vertical wall fires**.

Vertical wall fires represent a canonical configuration in fire safety engineering. The upward boundary layer flow generated by the burning fuel creates a strong feedback loop between heat release, buoyancy forces, and entrainment rates. Modeling these phenomena requires extending the numerical methodology to capture:

- **Reactive chemistry and heat release:** Multi-component species transport and combustion models.
- **Thermal radiation:** Spatial heat flux distribution onto the wall boundary.
- **Flame-wall interactions:** Near-wall momentum and thermal boundary layer behavior under high temperature gradients.

6.2 Physical Modelling

To solve the complex turbulent reacting flow within the vertical wall configuration, several physical sub-models are coupled to the governing transport equations. Specifically, the simulation incorporates a Large Eddy Simulation (LES) turbulence framework, a low-Mach hydrodynamic solver, an Eddy Dissipation Model (EDM) diffusion flame combustion model for propylene oxidation, a Finite Volume Method (FVM) radiation model for thermal feedback, and a fixed soot fraction representation to account for radiative emission and absorption across the plume as detailed in section 5 .

6.3 Benchmark Configuration & Reference Setup

To validate the reactive solver setup, the numerical domain and boundary conditions are adapted from the experimental and numerical benchmark study by Ren et al. [6].

Key Features of the Reference Configuration:

1. **Porous Burner Injection:** Gaseous fuel is uniformly injected normal to a vertical wall surface through single or modular porous burner panels.
2. **Boundary Layer Growth:** As the gaseous fuel reacts with the surrounding entrained air, a turbulent boundary layer flame develops vertically along the wall face.
3. **Wall Thermal Response:** The solid wall surface experiences combined convective and radiative heat flux, creating strong temperature variations across the boundary layer.

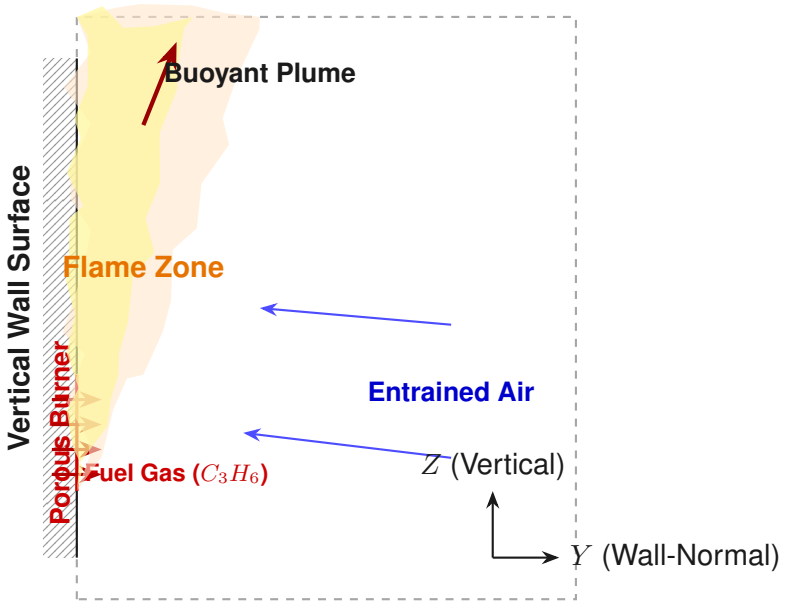


Figure 21. Schematic representation of the turbulent vertical wall fire configuration adapted from Ren et al. [6].

By building upon the grid spacing and numerical modeling strategies validated during the DHVC sensitivity analysis, this section assesses the solver's performance in predicting key turbulent wall fire metrics, including vertical velocity distributions, temperature profiles, near-wall grid requirements (y^+), and wall-law applicability in reactive flow regimes.

6.4 Fluid and Physical Parameters

The primary fluid properties, thermal conditions, and fuel injection parameters for the reactive vertical wall simulation are summarized in Table 11.

Table 11. Fluid properties and physical parameters for the vertical wall fire simulation.

Parameter	Symbol	Value & Unit
Ambient Temperature	T_0	$300.0\,\text{K}$
Wall / Fuel Temperature	T_f	$348.0\,\text{K}$
Prandtl Number	Pr	0.7
Reference Dynamic Viscosity	μ_{ref}	$1.68 \times 10^{-5}\,\text{kg/(m}\cdot\text{s)}$
Propylene Mass Flux	$\dot{m}''$	$0.0171\,\text{kg/(m}^2\cdot\text{s)}$
Burner Surface Area	A_{burner}	$0.38\,\text{m}^2$
Total Fuel Mass Flow Rate	$\dot{m}$	$0.0065\,\text{kg/s}$

The reference dynamic viscosity μ_{ref} is evaluated at the ambient reference temperature $T_0 = 300.0$ K, with temperature-dependent variations modeled using Sutherland's law:

$$\mu(T) = \mu_{\text{ref}} \left(\frac{T}{T_0} \right)^{3/2} \frac{T_0 + S}{T + S} \tag{45}$$

where S represents Sutherland's constant specific to the fluid medium.

6.5 Geometry and Computational Domain

The physical configuration models a three-dimensional vertical wall fire, where gaseous propylene is injected through a wall-mounted burner to establish a turbulent buoyant plume along a vertical boundary. The computational domain spans $X \in [0.0, 0.38]$ m in width, $Y \in [0.0, 0.36]$ m in wall-normal depth, and $Z \in [-0.06, 1.46]$ m in vertical height, yielding a total domain height of 1.52 m. The fuel-injection burner face is flush with the wall at $Y = 0.0$ m and occupies the vertical span $Z \in [0.0, 1.0]$ m.

Table 12. Computational Domain Dimensions and Spatial Discretization Strategy.

Axis	Physical Coordinate	Extent [m]	Discretization Strategy
Width (X)	$X \in [0.0, 0.38]$	0.38	Uniform spacing ($\delta_0 = 10$ mm)
Depth (Y)	$Y \in [0.0, 0.36]$	0.36	Multi-zone stretched grid (δy_{wall} to 10 mm)
Height (Z)	$Z \in [-0.06, 1.46]$	1.52	Uniform spacing ($\delta_0 = 10$ mm)

6.6 Boundary Conditions and Active Scalar Formulations

The boundary conditions imposed across the domain faces are configured to accurately represent the physical wall-fired burner setup. In addition to the primary hydrodynamic and thermal fields, several active transport scalars are solved throughout the domain:

- **Fuel Species (C_3H_6):** Mass fraction tracking the injected propylene from the burner inlet.
- **Oxidizer (O_2) and Inert (N_2):** Mass fractions defining the entrained ambient air composition.
- **Combustion Products (CO_2 and H_2O):** Scalar fields tracking the generated gaseous products of the reaction.
- **Soot Mass Fraction (Y_s):** Active scalar representing the soot distribution ($Y_s = 0.095$) coupled with the radiation and combustion routines.

The specific numerical values, scalar fields, and governing configurations applied to each domain boundary type are defined as follows:

- **Burner Inflow Boundary ($Y = 0.0$ m, $Z \in [0.0, 1.0]$ m):** A mass-inflow boundary specifying a constant propylene mass flux of 0.0171 kg/(m$^2 \cdot$s). The inflow temperature is explicitly set to $T_f = 348$ K, with incoming radiation evaluated at this same temperature. Active scalar transport equations for the fuel ensure full purity injection, prescribing a mixture fraction and fuel mass fraction of $Z = 1.0$ and $Y_F = 1.0$, respectively.
- **Solid Wall Boundaries ($Z = -0.06$ m and non-burner walls):** A strict no-slip condition is enforced for the velocity field ($\mathbf{v} = 0$). The thermal boundary condition and radiative surface temperature are conditionally prescribed based on the height coordinate Z: an ambient temperature baseline of $T_o = 300$ K for $Z \leq 0$, transitioning to the elevated burner temperature $T_f = 348$ K for $Z > 0$. Radiative intensity is modeled using a gray surface assumption with a surface emissivity of 1.0, and standard zero-gradient conditions are applied to species mass fractions.
- **Open Entrainment Boundaries (Top $Z = 1.46$ m, front $Y = 0.36$ m, and lower non-burner surfaces):** Operating as an inlet/outlet boundary exposed to ambient conditions, the temperature is set to the ambient reference $T_o = 300$ K. Incoming fuel and mixture fractions are maintained at zero ($Y_F = 0.0$, $Z = 0.0$) to represent clean ambient air entrainment (tracking O_2 and N_2), and incoming radiation is bounded by T_o.

- **Lateral Symmetry Boundaries ($X = 0.0$ m and $X = 0.38$ m):** Standard symmetry planes are applied to truncate the computational domain efficiently along the lateral axis.

6.7 Structured Meshing Strategy

The domain is discretized using a Cartesian structured grid solver . While the width (X) and height (Z) axes employ uniform grid spacing of $\delta_0 = 10$ mm, the wall-normal axis (Y) utilizes a three-zone stretch function to efficiently resolve near-wall boundary layer physics:

1. **Zone 1 — Near-Wall Sublayer ($0.0 \leq Y \leq 0.036$ m):** High-density uniform grid zone with a constant cell thickness of δy_{wall} to capture steep thermal and velocity gradients.
2. **Zone 2 — Boundary Layer Transition ($0.036 < Y \leq 0.12$ m):** Stretched mesh region smoothly expanding cell resolution from δy_{wall} up to the far-field spacing $\delta_0 = 10$ mm.
3. **Zone 3 — Far-Field Plume Domain ($0.12 < Y \leq 0.36$ m):** Uniform coarse grid with element thickness maintained at $\delta_0 = 10$ mm out to the entrainment boundary.

6.8 Grid Sensitivity and Convergence Criteria

To evaluate spatial grid independence and ensure adequate resolution within the viscous and thermal sublayers (y^+ grid verification), four distinct near-wall grid spacings (δy_{wall}) were systematically analyzed:

- **Coarse Mesh:** Baseline wall-normal cell thickness set to $\delta y_{\text{wall}} = 5.0$ mm.
- **Intermediate Mesh:** Refined wall-normal cell thickness set to $\delta y_{\text{wall}} = 2.0$ mm.
- **Fine Mesh:** High-resolution wall-normal cell thickness set to $\delta y_{\text{wall}} = 1.0$ mm.
- **Finer Mesh:** Ultra-high resolution wall-normal cell thickness set to $\delta y_{\text{wall}} = 0.5$ mm.

Comparative evaluation of flow fields, thermal structures, and turbulent fluctuation profiles across these four refinement levels establishes spatial grid convergence and isolates near-wall discretization errors.

6.9 Grid Convergence and Boundary Layer Dynamics

6.9.1 Mean Velocity and Temperature Profiles

Wall-normal profiles of mean vertical velocity (W) and mean temperature (T) evaluated at $Z = 0.77$ m demonstrate grid sensitivity and numerical convergence behavior across all four grid resolution levels:

- **Mean Vertical Velocity (W):** The buoyant jet core exhibits a peak velocity ($W_{\text{max}} \approx 3.7 - 3.8$ m/s) near $Y \approx 0.015$ m. As the wall-normal spacing is refined from $\delta y_{\text{wall}} = 5.0$ mm down to 0.5 mm, profiles demonstrate asymptotic convergence toward the reference LES data, accurately capturing both the shear layer peak and the outer ambient air entrainment region.
- **Mean Temperature (T) & Flame Structure:** Wall-normal refinement from 5.0 mm to 0.5 mm resolves steeper thermal gradients near the wall, shifting the peak temperature ($T \approx 1150$ K) inward toward $Y \approx 0.018$ m. Finer grids ($\delta y_{\text{wall}} \leq 1.0$ mm) effectively eliminate coarse-grid over-broadening of the thermal boundary layer.
- **Thermal Core Discrepancy:** The peak mean temperature remains slightly underpredicted compared to the reference LES benchmark (~ 1250 K). This local attenuation is primarily

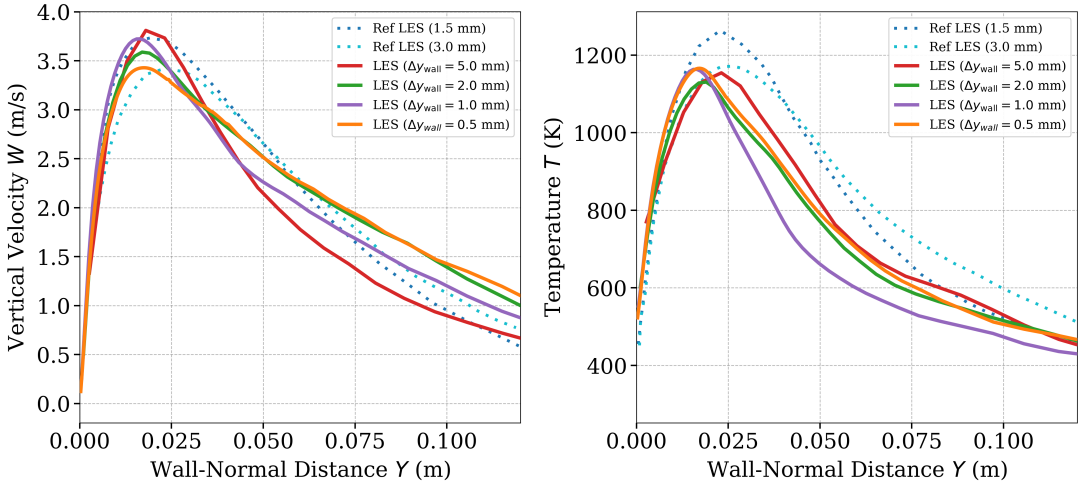


Figure 22. Comparison of mean vertical velocity W (left) and mean temperature T (right) profiles along wall-normal distance Y at height $Z = 0.77$ m across four grid refinement levels ($\delta y_{\text{wall}} = 5.0, 2.0, 1.0, 0.5$ mm) against reference LES data.

driven by numerical dissipation within the scalar convection scheme, which dampens unresolved scalar variance near the intense chemical reaction zone.

6.9.2 Turbulent Fluctuation Profiles

Time-averaged root-mean-square (RMS) profiles at $Z = 0.77$ m highlight the differential performance of the scalar fluctuation fields across the four mesh configurations:

- **Vertical Velocity Fluctuations (w'_{rms}):** Shows excellent agreement with reference LES profiles across all grid configurations, peaking at $w'_{\text{rms}} \approx 1.45$ m/s around $Y \approx 0.015 - 0.02$ m. The low-dissipation centered momentum discretization preserves vertical turbulent kinetic energy and vortex shedding dynamics effectively.
- **Temperature Fluctuations (T'_{rms}):** Peak variance is underpredicted across all cases ($\sim 380 - 400$ K peak vs. ~ 520 K reference). While mesh refinement down to $\delta y_{\text{wall}} = 0.5$ mm sharpens the profile gradient and shifts the peak towards the wall, scalar dissipation inherent in the transport formulation limits local variance growth.
- **Transverse Fluctuations (v'_{rms} and u'_{rms}):** Wall-normal fluctuations ($v'_{\text{rms}} \approx 0.45 - 0.50$ m/s) are well-constrained by kinematic wall-blocking. Conversely, streamwise fluctuations (u'_{rms}) display stronger sensitivity to grid resolution, where coarse grids ($\delta y_{\text{wall}} = 5.0$ mm) underpredict peak variance (~ 0.25 m/s) compared to fine grid levels (~ 0.35 m/s).

6.9.3 Inner-Unit (y^+) Scaled Boundary Layer Dynamics

Transforming wall-normal profiles into viscous wall units (y^+) confirms structural convergence within the conductive sublayer, buffer region, and turbulent outer plume:

- **Sublayer Resolution & Cell Centroids:** Refining δy_{wall} from 5.0 mm to 0.5 mm shifts the first off-wall cell centroid into the viscous sublayer ($y^+ \leq 1$), preventing numerical distortion of near-wall velocity and thermal gradients.
- **Mean Vertical Velocity (V_z vs. y^+):** Demonstrates consistent collapse in the linear sublayer ($y^+ < 10$). The buoyant velocity peak shifts from $y^+ \approx 20$ on coarse grids to $y^+ \approx 10^2$ on fine grids ($\delta y_{\text{wall}} \leq 1.0$ mm), capturing the correct buffer layer shear stress layer.

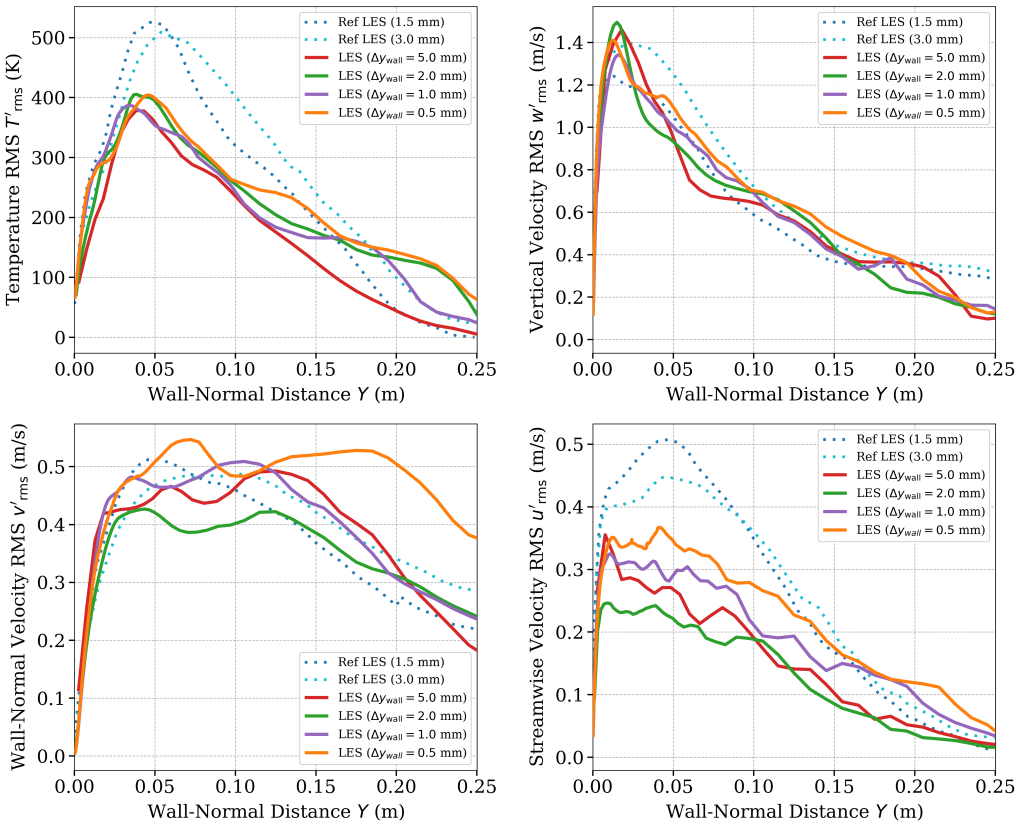


Figure 23. Time-averaged RMS velocity components ($w'_{\text{rms}}, v'_{\text{rms}}, u'_{\text{rms}}$) and temperature fluctuation (T'_{rms}) profiles at $Z = 0.77$ m for four grid refinement levels.

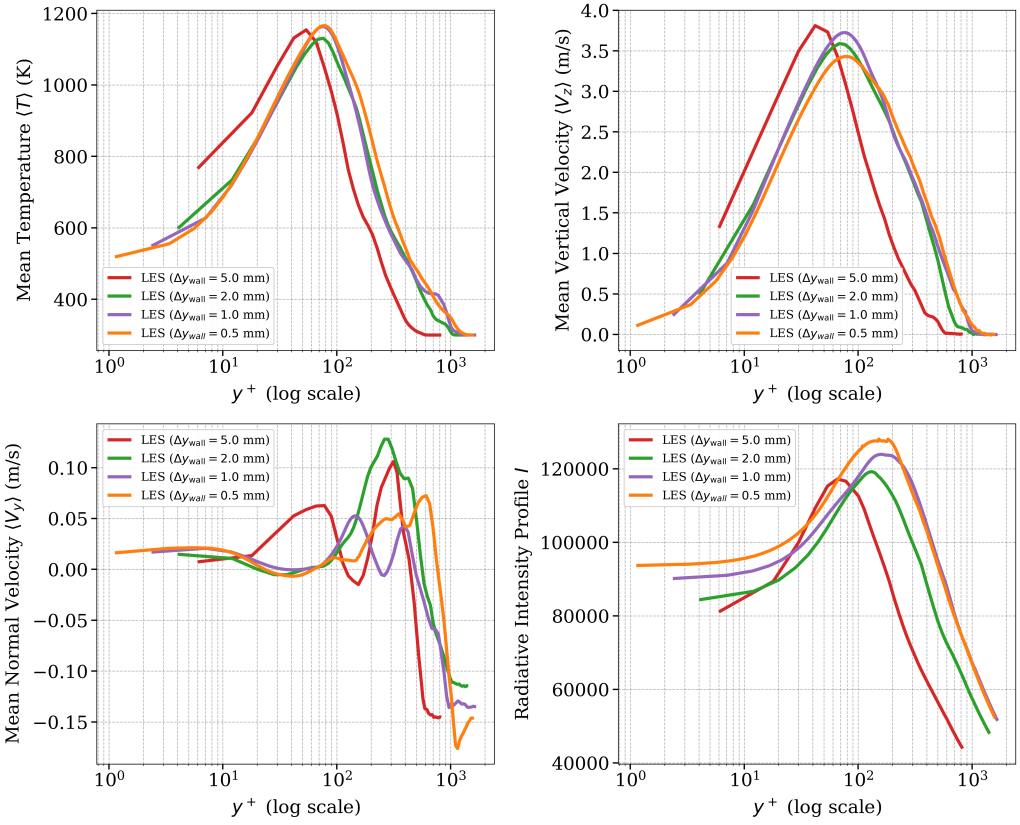


Figure 24. Mean flow, thermal, and radiative intensity profiles plotted against viscous wall units (y^+) on a logarithmic scale for grid resolutions $\delta y_{\text{wall}} \in [0.5, 5.0]$ mm.

- **Thermal Sublayer (T vs. y^+):** Grid refinement sharpens the conductive sublayer slope ($y^+ < 10$), placing the thermal peak precisely at $y^+ \approx 60 - 80$. This resolution is essential for accurately capturing parietal convective heat flux gradients.But the temperature captured near the wall is above the injected wall temperature in all cases converging to a higher value and further test using the Inlet condition for the burner is being tested to see its behaviour and to get a better understanding of the situation
- **Wall-Normal Velocity & Radiative Intensity (V_y and I vs. y^+):** The normal velocity (V_y) captures negative entrainment fluxes near the wall before transitioning to boundary layer expansion ($y^+ > 10^2$). The radiative intensity profile (I) reaches maximum emission at $y^+ \approx 10^2$, matching the high-temperature reaction zone core.

6.10 Wall Heat Flux Analysis

6.10.1 Convective Heat Flux (q''_{conv})

Wall-normal temperature gradients at the solid boundary govern convective heat transfer along the height Z of the vertical wall:

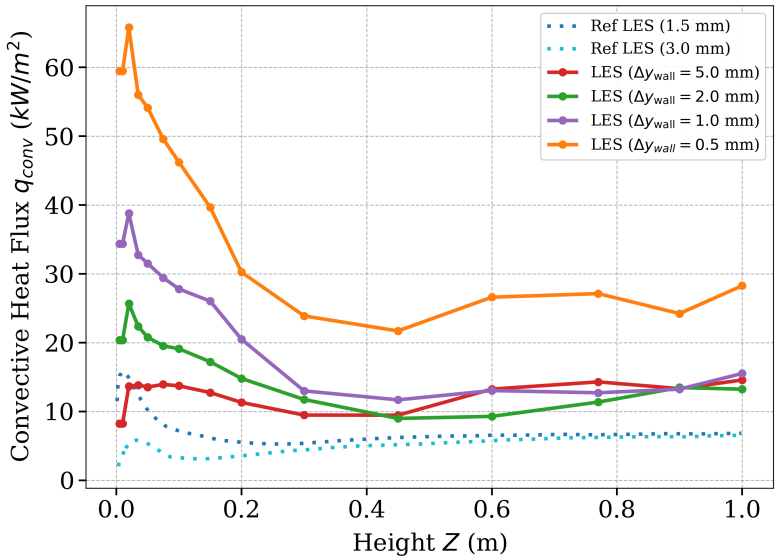


Figure 25. Distribution of convective heat flux q''_{conv} along wall height Z across four grid resolution levels ($\delta y_{\text{wall}} = 5.0, 2.0, 1.0, 0.5$ mm) compared against reference LES datasets.

- **Near-Wall Peak ($Z < 0.2$ m):** Grid refinement significantly elevates the convective peak near the burner base ($Z \approx 0.01$–0.02 m). As grid spacing is reduced, the captured peak flux increases from ~ 14 kW/m^2 (5.0 mm) and ~ 26 kW/m^2 (2.0 mm) to ~ 39 kW/m^2 (1.0 mm), reaching ~ 67 kW/m^2 on the finest grid ($\delta y_{\text{wall}} = 0.5$ mm). This increase is directly driven by resolving the conductive boundary layer gradient ($\partial T/\partial y|_{\text{wall}}$).
- **Downstream Behavior ($Z > 0.4$ m):** In the fully developed region, convective flux for the 5.0 mm, 2.0 mm, and 1.0 mm grids stabilizes into a narrow band of 10–16 kW/m^2. The ultra-fine mesh (0.5 mm) stabilizes at a higher baseline (~ 25–28 kW/m^2), reflecting enhanced resolution of small-scale turbulent shear structures along the wall. But the convective profile seems not to be converged , as seen from the temperature profile against y^+ the wall temperature is not correctly captured so this answers the non convergence of the convective profile and the Inlet condition is tested to address this issue.

6.10.2 Radiative Heat Flux (q''_{rad})

Radiative emission from hot gas combustion products and soot dictates the radiative flux received by the wall surface:

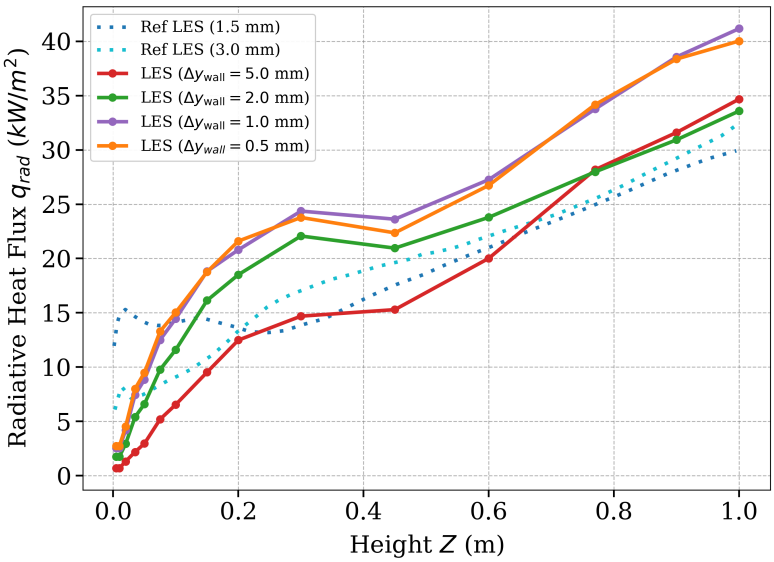


Figure 26. Distribution of radiative heat flux q''_{rad} along wall height Z across four grid resolution levels and reference LES cases.

- **Monotonic Downstream Increase:** Radiative heat flux increases continuously along the wall height ($Z = 0$ to $1.0\,\text{m}$), growing from near zero at the burner origin to 33–$41\,\text{kW/m}^2$ at the top as the thermal plume thickness and optical path length expand.
- **Grid Sensitivity & Asymptotic Convergence:** Coarser grids ($\delta y_{\text{wall}} = 5.0\,\text{mm}$ and $2.0\,\text{mm}$) predict lower downstream radiative fluxes (~ 33–$34\,\text{kW/m}^2$ at $Z = 1.0\,\text{m}$). The finer grids ($\delta y_{\text{wall}} = 1.0\,\text{mm}$ and $0.5\,\text{mm}$) show strong asymptotic convergence, collapsing onto virtually identical profiles that reach ~ 40–$41\,\text{kW/m}^2$ downstream due to accurately resolved peak temperatures in the reaction core.

6.10.3 Total Heat Flux (q''_{tot}) and Experimental Validation

The total wall heat flux ($q''_{\text{tot}} = q''_{\text{conv}} + q''_{\text{rad}}$) is validated against digitized experimental data from de Ris et al. [5]:

- **Flow Regime Transition:** A vertical line at $Z = 0.3\,\text{m}$ marks the threshold where the buoyant boundary layer transitions into a fully turbulent state.
- **Experimental Agreement:** Experimental measurements range from ~ 10–$15\,\text{kW/m}^2$ near the base up to $\sim 37\,\text{kW/m}^2$ at $Z = 1.0\,\text{m}$. The coarse mesh ($\delta y_{\text{wall}} = 5.0\,\text{mm}$) closely matches experimental data points in the lower and intermediate height regions ($Z \leq 0.5\,\text{m}$) where under-resolved convective gradients offset scalar dissipation errors.
- **High-Resolution Behavior:** Finer grid spacings ($\delta y_{\text{wall}} \leq 1.0\,\text{mm}$) yield higher total flux values across the domain, driven by the intense convective peak at the burner base ($q''_{\text{tot}} \approx 68\,\text{kW/m}^2$ for $0.5\,\text{mm}$) and the additive superposition of converged radiative emission downstream ($q''_{\text{tot}} \approx 58$–$68\,\text{kW/m}^2$ at $Z = 1.0\,\text{m}$).

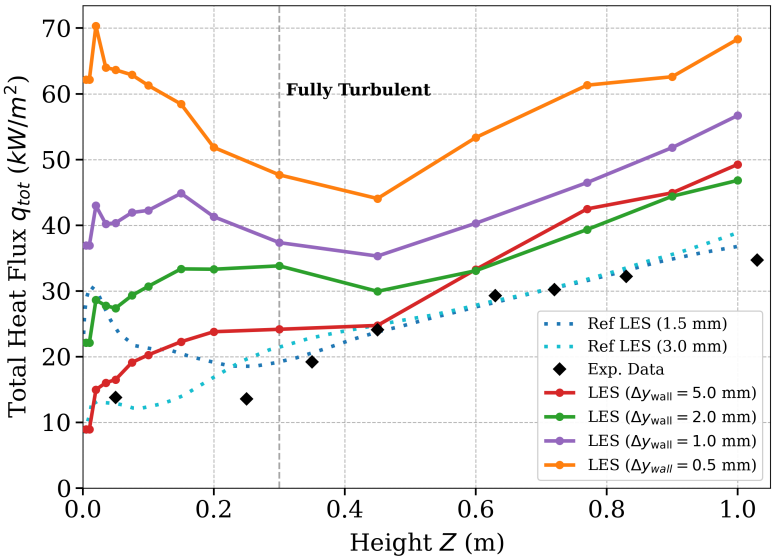


Figure 27. Total heat flux q''_{tot} profile along height Z compared against experimental measurement data from de Ris et al. [5] and reference LES results.

6.11 Quantitative Mesh Quality and Performance Metrics

A quantitative comparison of the wall-resolution metrics, non-dimensional heat transfer parameters, and LES resolution quality criteria across the four grid refinement levels is summarized in Table 13.

Table 13. Summary of near-wall grid resolution (y^+), skin friction coefficient (C_f), Nusselt number (Nu), and LES quality criteria (Pope and Celik) for the vertical wall simulations.

Simulation	Δy_{wall} (m)	LES Model	y^+	C_f	Nu	Pope	Celik (%)
1	0.0050	Dynamic Smagorinsky	14.43	0.00092	153.70	0.9471	18.43
2	0.0020	Dynamic Smagorinsky	5.47	0.00087	237.15	0.9564	8.42
3	0.0010	Dynamic Smagorinsky	2.68	0.00082	365.75	0.9459	5.28
4	0.0005	Dynamic Smagorinsky	1.27	0.00069	544.52	0.9517	2.98

6.11.1 Key Observations

- **Near-Wall Resolution (y^+):** Refining Δy_{wall} from $0.0050\,\text{m}$ ($5.0\,\text{mm}$) down to $0.0005\,\text{m}$ ($0.5\,\text{mm}$) systematically reduces the first off-wall cell centroid from $y^+ = 14.43$ to $y^+ = 1.27$, placing the grid strictly inside the viscous sublayer ($y^+ \approx 1$).
- **Convective Heat Transfer (Nu):** While the Nusselt number increases with grid refinement from 153.70 to 544.52, the near-wall temperature profile fails to accurately recover the prescribed wall temperature ($T_w = 348$ K), instead overshooting above 500 K in the first cell. This discrepancy indicates that the steep gradients are influenced by thermal coupling constraints and the propagation of the inflow condition rather than achieving the true asymptotic wall limit solely through grid resolution.
- **Skin Friction (C_f):** Shows a gradual reduction ($0.00092 \rightarrow 0.00069$), stabilizing as velocity shear gradients near the boundary layer are fully resolved.
- **LES Resolution Quality (Pope & Celik):**

– **Pope Criterion:** Values remain consistently above 0.945 across all grid configurations (0.9459–0.9564), confirming that over 94% of the turbulent kinetic energy is directly resolved by the grid.

– **Celik Index:** Decreases substantially from 18.43% on the coarse mesh down to 2.98% on the finest mesh (0.5 mm), confirming a significant reduction in subgrid-scale discretization error.

6.12 Parametric Sensitivity: Influence of Inlet vs. Inflow Boundary Conditions

This section evaluates the sensitivity of the simulation to the specification of boundary conditions at the injection wall, comparing the performance of an *Inflow* condition against an explicit *Inlet* condition (both evaluated at a near-wall resolution of $\Delta y_{\text{wall}} = 0.5$ mm). In the inlet configuration, the injected velocity is explicitly modeled using the mass flux of propylene divided by the density of propylene ($u_{\text{inj}} = \dot{m}/\rho$). All other core numerical and physical parameters remain identical to the baseline configuration to isolate the impact of the boundary formulation.

6.12.1 Analysis of Mean Flow Fields and Transport Behavior

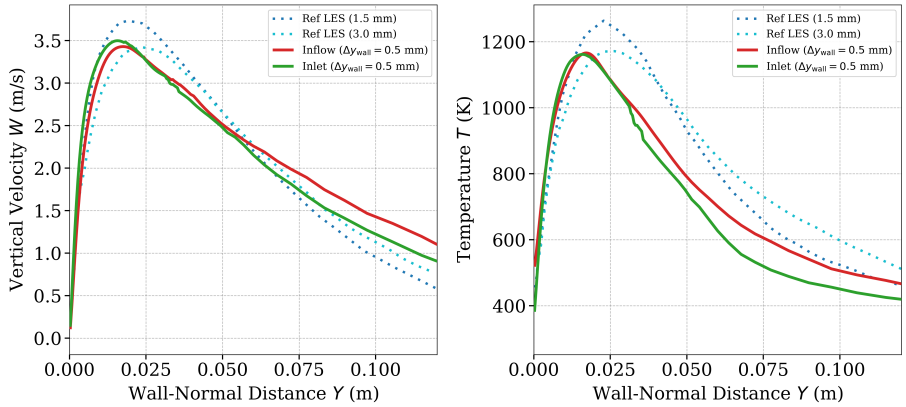


Figure 28. Comparison of mean vertical velocity profiles (W^*) and mean temperature profiles (T^*) across the normalized channel width for Inflow and Inlet boundary condition configurations against reference LES benchmarks.

The mean vertical velocity (W^*) and temperature (T^*) profiles presented in Figure 28 demonstrate that the **Inlet condition** captures the injected temperature at the wall with significantly higher fidelity than the Inflow condition. This accurate thermal capture at the boundary directly results in an improved prediction of the core convective profile and total heat flux.

While the Inflow condition handles overall mass flow smoothly (exhibiting flux independence in the core region), it introduces minor discrepancies in tracking sharp near-wall gradients. As highlighted by the distinct wall flux behavior ($\Phi_w \neq \Phi_{\text{wo}}$) unique to the Inlet condition, explicitly resolving the inlet mass-flux boundary condition allows for a more rigorous representation of local thermal injection, preventing the artificial damping observed in standard inflow setups.

6.12.2 Turbulence Statistics and RMS Profiles Across Boundary Conditions

Temperature and Velocity Fluctuations: The RMS profiles displayed in Figure 29 show that both boundary conditions successfully resolve turbulent fluctuations within acceptable bounds. However, the Inlet condition maintains a sharper, more physically consistent peak in temperature RMS near the wall boundary layers ($y \approx 0.025$ m), closely mirroring the reference LES data.

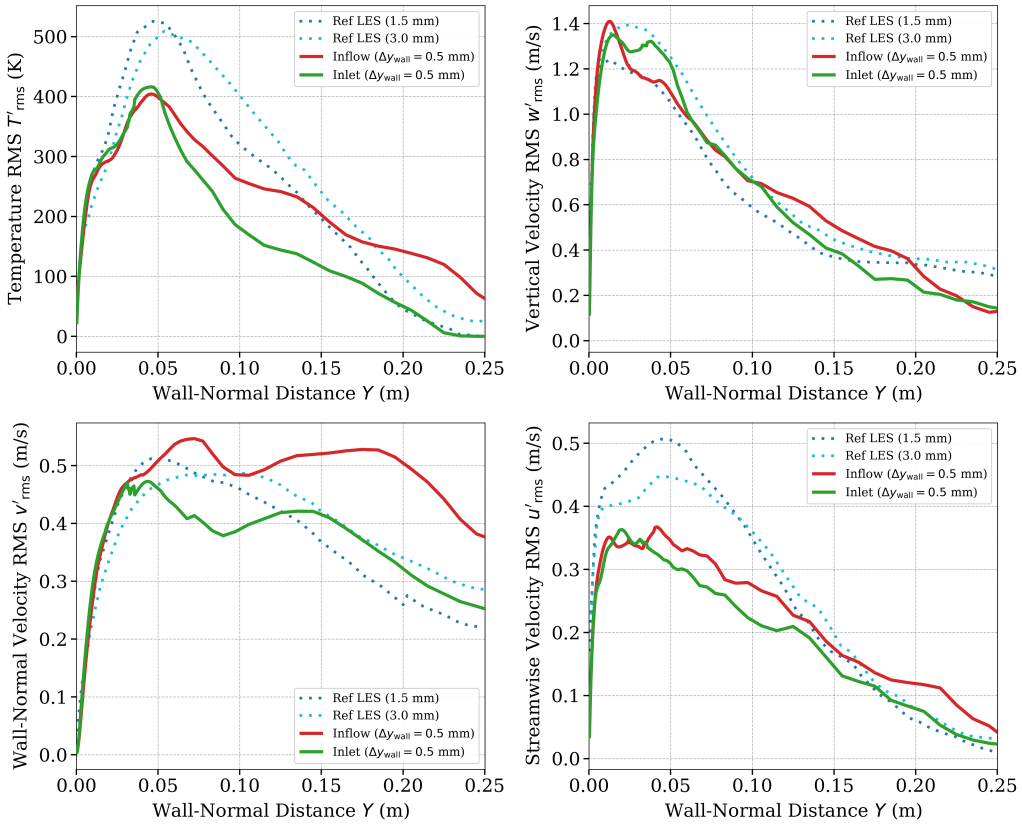


Figure 29. Comparison of RMS turbulence statistics across the wall-normal distance (y): (a) Temperature RMS, (b) Vertical velocity RMS (w'_{rms}), (c) Spanwise velocity RMS (v'_{rms}), and (d) Streamwise velocity RMS (u'_{rms}).

6.12.3 Inner Unit (y^+) Scaled Boundary Layer Dynamics

The log-scale y^+ evaluations in Figure 30 further confirm the advantage of the Inlet boundary formulation. The radiative intensity profile and mean normal velocity components demonstrate that the Inlet condition properly resolves the near-wall scalar and flux contributions without generating unphysical oscillations in the buffer layer.The mean temperature profile shows that the wall temperature is well captured by the Inlet condition rather than the inflow and it address our problem

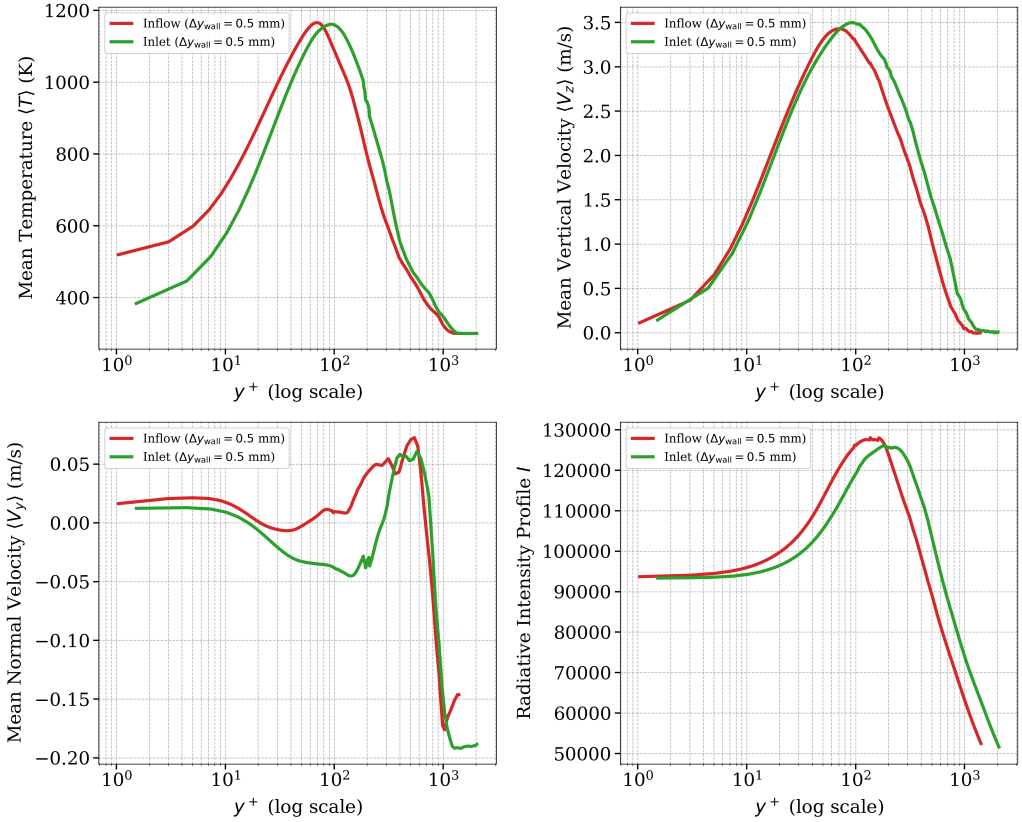


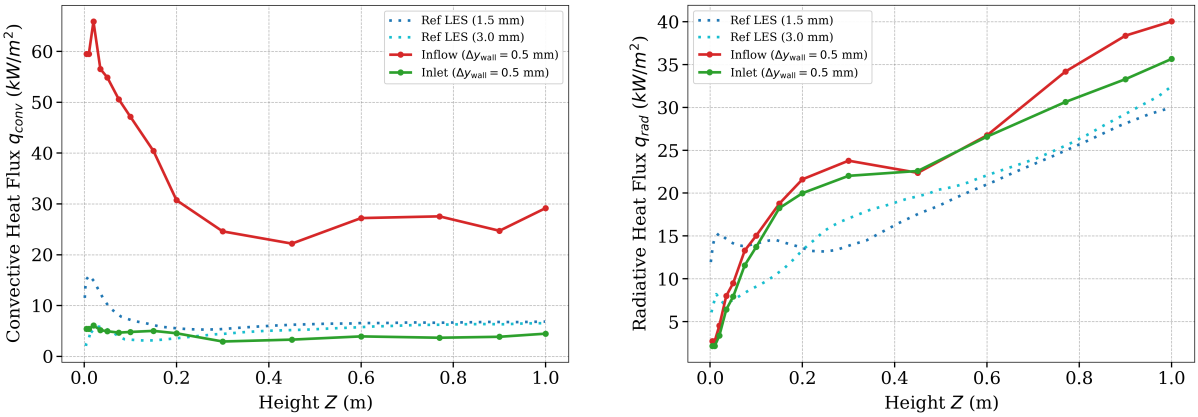
Figure 30. Profiles plotted against the log-scale wall coordinate (y^+): (a) Mean temperature, (b) Mean vertical velocity, (c) Mean normal velocity, and (d) Radiative intensity profile.

6.13 Analysis of Heat Flux Distribution Across Boundary Conditions

Convective Heat Flux Distribution: The spatial distribution of the convective heat flux (q_{conv}) along the channel height (Z) is displayed in Figure 31a. The Inlet condition successfully tracks the reference LES benchmarks, maintaining a low, stable convective heat flux profile ($\approx 0 - 5$ kW/m^2) across the entire height. In contrast, the standard Inflow condition exhibits a severe overprediction near the lower entrance region ($Z < 0.2$ m), spiking up to over 60 kW/m^2 before gradually stabilizing. This highlights how explicitly resolving the mass-flux-based inlet velocity prevents unphysical thermal energy accumulation near the boundaries.

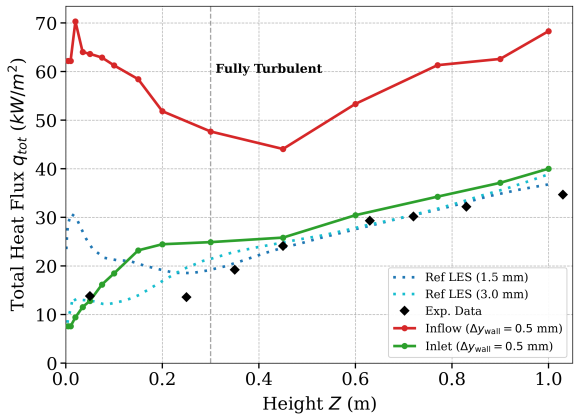
Radiative Heat Flux Distribution: The radiative heat flux (q_{rad}) profiles shown in Figure 31b demonstrate consistent behavior between both boundary configurations. Both the Inflow and Inlet setups follow the overall upward trend of the reference LES data along the height of the channel (Z), with the Inlet condition providing a slightly smoother alignment with the 1.5 mm and 3.0 mm reference benchmarks in the core region.

Total Heat Flux and Experimental Comparison: The total heat flux (q_{tot}) profiles, combining both convective and radiative contributions, are illustrated in Figure 31c alongside experimental



(a) Convective Heat Flux (q_{conv})

(b) Radiative Heat Flux (q_{rad})



(c) Total Heat Flux (q_{tot})

Figure 31. Comparison of heat flux distributions along the channel height (Z): (a) Convective heat flux, (b) Radiative heat flux, and (c) Total heat flux against reference LES benchmarks and experimental data points.

data points. Because the convective component is heavily over-predicted by the Inflow condition, its total heat flux curve suffers from massive artificial inflation at the entrance. Conversely, the Inlet condition closely matches the reference LES curves and aligns remarkably well with the experimental data markers across the fully turbulent zone ($Z > 0.3$ m), proving that proper boundary specification is essential for accurate thermal predictions.

6.14 Inference and Perspective: Boundary Formulation and Flux Dynamics

The comparative evaluation between the generalized *Inflow* and explicit *Inlet* boundary conditions highlights a critical balance required in natural convection modeling. While neither condition is universally flawed across all domains, each excels in different physical regimes:

- **Core Transport vs. Near-Wall Accuracy:** The standard *Inflow* condition handles overall core mass flow and macro-transport smoothly, maintaining relative flux independence in the channel center. However, it fails to capture sharp near-wall gradients accurately.
- **The Role of Wall Flux Coupling ($\Phi_w \neq \Phi_{w0}$):** The explicit *Inlet* condition introduces a rigorous coupling based on the mass injection rate of propylene relative to density ($u_{\text{inj}} = \dot{m}/\rho$). This formulation correctly modifies the wall flux ($\Phi_w \neq \Phi_{w0}$), successfully suppressing the massive artificial overpredictions of convective heat flux (q_{conv}) seen in standard inflow setups.

- **Achieving a Physical Balance:** Consequently, while Inflow provides stable core behavior, the Inlet condition is essential for resolving the local thermal boundary layer. Balancing these aspects shows that proper boundary specification must integrate both core mass-flux consistency and rigorous near-wall scalar coupling to match experimental total heat flux (q_{tot}) data.

Ultimately, this analysis demonstrates that a successful LES framework must strike a balance: utilizing mass-flux-based inlet formulations to accurately govern local wall injection while preserving the stable core transport characteristics of the flow.

7 Conclusion of Turbulent Wall Fire Case

The objective of this work was to establish the numerical resolution and modelling requirements for extending the LES methodology from canonical buoyancy-driven flows towards the simulation of fire propagation over solid combustible surfaces. Following the validation of the non-reactive DHVC configuration, the methodology was applied to a reactive, wall-bounded turbulent fire to identify the main mesh, modelling and boundary-condition parameters required to accurately capture the flow, thermal field and wall heat transfer. The reactive wall-fire study confirmed the importance of the wall-normal resolution for resolving the near-wall thermal and velocity gradients. The adopted three-zone wall-normal meshing strategy, consisting of a fine near-wall region, a stretched transition region and a coarser outer region, allows the computational resolution to be concentrated where it is physically required. Among the tested configurations, a first wall-normal cell size of **0.5 mm** provided a first-cell position of approximately $y^+ = 1.27$ and a **Celik discretization index of 2.98%**, indicating an adequately resolved near-wall region. This resolution therefore provides a reference for the near-wall mesh in subsequent simulations, while maintaining a coarser resolution in the outer flow. The results also showed that the prediction of wall heat transfer is strongly linked to the resolution of the thermal boundary layer, particularly for the convective contribution. The comparison of the **Inflow** and **explicit Inlet** formulations further demonstrated the importance of an appropriate treatment of the fuel injection boundary. The Inflow condition did not correctly reproduce the near-wall temperature distribution, resulting in discrepancies in the predicted convective heat flux. In contrast, the explicit Inlet formulation based on the prescribed fuel mass flux provided a more consistent near-wall thermal field and should therefore be retained when accurate surface heat transfer is required. Together with the results obtained from the DHVC study, these findings define a numerical baseline for future solid-fuel fire-propagation simulations. This baseline consists of the **three-zone wall-normal mesh with a near-wall resolution of approximately 0.5 mm**, a coarser outer region, the **low-dissipation centred formulation for the momentum equations**, the LES subgrid-scale modelling strategy established during the validation study, and a **second-order implicit temporal discretization controlled by the CFL condition**. The explicit mass-flux-based Inlet formulation is also identified as an essential boundary treatment for accurately predicting the thermal interaction between the flame and the fuel surface. Overall, the study provides the numerical framework required to move from a validated turbulent wall-fire configuration towards a large-scale fire-propagation simulation. The identified mesh resolution, three-zone meshing strategy, numerical discretization, temporal treatment and fuel-injection boundary condition provide the basis for coupling the gas-phase flame dynamics with the heating and pyrolysis of a solid fuel plate. This framework can subsequently be applied to the CISCCO [7] configuration to investigate flame evolution and, ultimately, predict the **fire-propagation rate over the solid fuel surface**.

However, translating these validated parameters to full-scale CISCCO test conditions highlights major computational limitations. Estimating the numerical setup reveals that strict time-step constraints ($\Delta t \sim 10^{-4}$ s) combined with millions of grid cells result in a prohibitive CPU cost to

simulate a complete, fully-resolved test scenario. This indicates that a trail and error approach is intractable for full industrial campaigns, requiring alternative strategies such as reducing grid accuracy (near the wall and core sizes) to simulate degraded approximations of the whole scenario, or focusing high-fidelity resolution on selected, short-duration phases (e.g., transient windows of ~ 10 s). These considerations provide an essential operational perspective for bridging fundamental benchmarks with future large-scale safety assessments.

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Study of the resolution of mass and heat transfer for large-scale simulation of fire propagation on solid fuel plates

Accurately capturing turbulent combustion, thermal radiation, and boundary layer dynamics in wall-bounded reactive flows is vital for predicting flame spread over solid combustible surfaces. This study provides a systematic Large Eddy Simulation (LES) investigation to establish grid resolution demands and numerical fidelity standards required prior to modeling large-scale fire propagation across plates and slabs. The investigation progressed from a non-reactive Differentially Heated Vertical Cavity (DHVC) sensitivity matrix evaluated across a spectrum of Rayleigh numbers (Ra) to assess numerical efficiency, to reactive vertical wall fire simulations across four wall-normal grid refinements ($\Delta y_{\text{wall}} = 5.0\,\text{mm}, 2.0\,\text{mm}, 1.0\,\text{mm}, 0.5\,\text{mm}$). The 2nd-order centered Navier–Stokes momentum scheme proved highly efficient and non-dissipative for fluid mechanics, successfully capturing coherent vortex structures and velocity statistics (w'_{rms}). However, scalar transport suffered from numerical dissipation in the convection scheme, damping temperature fluctuations (T'_{rms}) on coarser meshes. Refining the wall-normal grid ($\Delta y_{\text{wall}} \leq 0.5\,\text{mm}, y^+ \approx 1.27$) minimized numerical diffusion, contracting the flame profile toward its physical width and reducing the Celik discretization error index below 3%. Furthermore, comparative analyses between generalized inflow and explicit inlet boundary formulations demonstrated that proper coupling of mass-flux-based wall conditions ($\Phi_w \neq \Phi_{w0}$) is essential to balance near-wall convection-diffusion regimes ($u\delta/D \approx 1$), successfully suppressing artificial convective heat flux spikes while maintaining stable core macro-transport. These findings establish clear grid resolution guidelines and boundary specification standards to optimize accuracy and computational budget in upcoming simulations of fire propagation across horizontal and inclined plates and slabs.

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